# SRI (k-inflation): N03LO

The spectra are evaluated at $-k_*\eta_* = 1$, through relative amplitude order $p = 3$. The Hubble-flow parameters are denoted by $\epsilon_j$ and $M_{\rm Pl}$ is the reduced Planck mass. Coefficients are generated numerically from the normalized mode equations, using the iterative algebra. The convention for the spectral derivatives is

$$\alpha_n^A = \frac{d^{n+1}\ln P_A}{d(\ln k)^{n+1}}, \qquad \alpha_0^s = n_s - 1, \qquad \alpha_0^t = n_t, \qquad r = P_t/P_s.$$

For k-inflation, $c_t = 1$. The scalar amplitude includes transport from the acoustic reference to the common tensor reference. The sound speed is $c_{s*} > 0$, with

$$\alpha_n^{c_s} = \frac{d^{n+1}\ln c_s}{d(\ln k)^{n+1}}.$$

# 1 Spectra

## 1.1 Scalar spectrum

$$P_s(k_*) = \frac{H_*^2}{8\pi^2 M_{\rm Pl}^2 \epsilon_{1*} c_{s*}} \left[\sum_{d=0}^{3} F_{s,d} + O(\mathrm{SR}^4)\right].$$

**Relative order 0.**

$$F_{s,0} = 1.0$$

**Relative order 1.**

$$\begin{aligned} F_{s,1} = {} & 0.72963715453852\epsilon_2 - 0.54072569092296\epsilon_1 \\ & - 1.2703628454615\alpha_0^{c_s} - 1.0\alpha_0^{c_s}\log(c_s) \\ & - 1.0\epsilon_2\log(c_s) - 2.0\epsilon_1\log(c_s) \end{aligned}$$

**Relative order 2.**

$$\begin{aligned} F_{s,2} = {} & 1.5402686460338\epsilon_1^2 + 0.49988573877771\epsilon_2^2 \\ & - 0.39567736285244\alpha_1^{c_s} - 0.22975141576082\left(\alpha_0^{c_s}\right)^2 \\ & + 0.5\left(\alpha_0^{c_s}\right)^2\log(c_s)^2 + 0.5\epsilon_2^2\log(c_s)^2 \\ & + 2.0\epsilon_1^2\log(c_s)^2 + 0.27036284546148\left(\alpha_0^{c_s}\right)^2\log(c_s) \\ & + 1.0192767657904\epsilon_1\epsilon_2 + 1.0809943369567\alpha_0^{c_s}\epsilon_1 \\ & + 0.14504832807052\epsilon_2\epsilon_3 - 0.5\alpha_1^{c_s}\log(c_s)^2 \\ & - 0.91854861815409\epsilon_1^2\log(c_s) - 0.72986567698311\alpha_0^{c_s}\epsilon_2 \\ & - 1.2703628454615\alpha_1^{c_s}\log(c_s) - 0.72963715453852\epsilon_2^2\log(c_s) \\ & + 1.0\alpha_0^{c_s}\epsilon_2\log(c_s)^2 + 1.0\epsilon_1\epsilon_2\log(c_s)^2 \\ & + 2.0\alpha_0^{c_s}\epsilon_1\log(c_s)^2 + 1.0814513818459\alpha_0^{c_s}\epsilon_1\log(c_s) \\ & + 0.72963715453852\epsilon_2\epsilon_3\log(c_s) - 0.5\epsilon_2\epsilon_3\log(c_s)^2 \\ & - 0.45927430907704\alpha_0^{c_s}\epsilon_2\log(c_s) \\ & - 2.459274309077\epsilon_1\epsilon_2\log(c_s) \end{aligned}$$

**Relative order 3.**

$$\begin{aligned} F_{s,3} = {} & 0.20072294544592\epsilon_2^3 + 1.6057835635673\epsilon_1^3 \\ & - 0.2199597209462\alpha_2^{c_s} - 0.028914209092607\left(\alpha_0^{c_s}\right)^3 \\ & + 4.2977813647689\epsilon_1^2\epsilon_2 + 1.1508618640566\alpha_1^{c_s}\epsilon_1 \\ & + 0.50011426122229\left(\alpha_0^{c_s}\right)^3\log(c_s) + 0.17975356917588\alpha_1^{c_s}\epsilon_2 \end{aligned}$$

$$
\begin{aligned}
&+ 0.36481857726926 \left(\alpha_0^{c_s}\right)^3 \log\left(c_s\right)^2 \\
&+ 0.36481857726926\epsilon_2^3 \log\left(c_s\right)^2 + 2.9185486181541\epsilon_1^3 \log\left(c_s\right)^2 \\
&+ 1.959600275274\epsilon_1\epsilon_2^2 + 0.030669313835717\epsilon_2\epsilon_3^2 \\
&- 0.39567736285244\alpha_2^{c_s} \log\left(c_s\right) - 1.0906092762856\alpha_0^{c_s}\alpha_1^{c_s} \\
&- 0.12746831820078\alpha_0^{c_s}\epsilon_2^2 - 0.25470811395696 \left(\alpha_0^{c_s}\right)^2 \epsilon_1 \\
&- 0.63518142273074\alpha_2^{c_s} \log\left(c_s\right)^2 - 0.32050476429836\alpha_0^{c_s}\epsilon_1^2 \\
&- 0.16666666666667\alpha_2^{c_s} \log\left(c_s\right)^3 \\
&- 0.16666666666667 \left(\alpha_0^{c_s}\right)^3 \log\left(c_s\right)^3 \\
&- 0.16666666666667\epsilon_2^3 \log\left(c_s\right)^3 - 1.3333333333333\epsilon_1^3 \log\left(c_s\right)^3 \\
&- 0.3571054727393 \left(\alpha_0^{c_s}\right)^2 \epsilon_2 - 0.23520055946694\epsilon_2^2\epsilon_3 \\
&- 0.49988573877771\epsilon_2^3 \log\left(c_s\right) - 3.9990859102216\epsilon_1^3 \log\left(c_s\right) \\
&+ 0.5\alpha_0^{c_s}\alpha_1^{c_s} \log\left(c_s\right)^3 + 0.5\alpha_1^{c_s}\epsilon_2 \log\left(c_s\right)^3 \\
&+ 0.5\epsilon_2^2\epsilon_3 \log\left(c_s\right)^3 + 1.0\alpha_1^{c_s}\epsilon_1 \log\left(c_s\right)^3 \\
&+ 3.0\epsilon_1^2\epsilon_2 \log\left(c_s\right)^2 \\
&+ 0.50034278366688 \left(\alpha_0^{c_s}\right)^2 \epsilon_2 \log\left(c_s\right) \\
&+ 2.2898599312799\alpha_0^{c_s}\epsilon_1\epsilon_2 \\
&+ 1.1889114636156 \left(\alpha_0^{c_s}\right)^2 \epsilon_1 \log\left(c_s\right)^2 \\
&+ 2.3778229272311\alpha_0^{c_s}\epsilon_1^2 \log\left(c_s\right)^2 \\
&+ 1.8110885363844\alpha_1^{c_s}\epsilon_1 \log\left(c_s\right)^2 + 0.49443659507158\alpha_0^{c_s}\epsilon_2\epsilon_3 \\
&+ 1.0944557318078\alpha_0^{c_s}\epsilon_2^2 \log\left(c_s\right)^2 \\
&+ 1.0944557318078 \left(\alpha_0^{c_s}\right)^2 \epsilon_2 \log\left(c_s\right)^2 \\
&+ 0.14504832807052\epsilon_2\epsilon_3^2 \log\left(c_s\right) \\
&+ 1.2296371545385\epsilon_1\epsilon_2^2 \log\left(c_s\right)^2 \\
&+ 0.36481857726926\epsilon_2\epsilon_3^2 \log\left(c_s\right)^2 \\
&+ 0.40554426819222\alpha_0^{c_s}\alpha_1^{c_s} \log\left(c_s\right)^2 \\
&+ 0.40554426819222\alpha_1^{c_s}\epsilon_2 \log\left(c_s\right)^2 \\
&+ 0.85472314948489\epsilon_2^2\epsilon_3 \log\left(c_s\right) \\
&+ 0.45995987641081 \left(\alpha_0^{c_s}\right)^2 \epsilon_1 \log\left(c_s\right) \\
&+ 1.8723490626616\alpha_1^{c_s}\epsilon_1 \log\left(c_s\right) + 0.030669313835717\epsilon_2\epsilon_3\epsilon_4 \\
&- 0.5\alpha_0^{c_s}\epsilon_2^2 \log\left(c_s\right)^3 \\
&- 0.5 \left(\alpha_0^{c_s}\right)^2 \epsilon_2 \log\left(c_s\right)^3 \\
&- 1.0 \left(\alpha_0^{c_s}\right)^2 \epsilon_1 \log\left(c_s\right)^3 \\
&- 2.0\alpha_0^{c_s}\epsilon_1^2 \log\left(c_s\right)^3 \\
&- 0.33418831413068\alpha_1^{c_s}\epsilon_2 \log\left(c_s\right) \\
&- 4.957559194624\epsilon_1^2\epsilon_2 \log\left(c_s\right) \\
&- 2.0800802471784\alpha_0^{c_s}\epsilon_1^2 \log\left(c_s\right) \\
&- 1.0944557318078\epsilon_2^2\epsilon_3 \log\left(c_s\right)^2 - 0.65969152604436\epsilon_1\epsilon_2\epsilon_3 \\
&- 0.16666666666667\epsilon_2\epsilon_3^2 \log\left(c_s\right)^3 \\
&- 0.33333333333333\epsilon_1\epsilon_2^2 \log\left(c_s\right)^3 \\
&- 1.3341883141307\alpha_0^{c_s}\alpha_1^{c_s} \log\left(c_s\right) \\
&- 2.7294086320939\epsilon_1\epsilon_2^2 \log\left(c_s\right)
\end{aligned}
$$

$$
\begin{aligned}
&- 0.49965721633312\alpha_0^{c_s}\epsilon_2^2\log(c_s)\\
&+ 0.5\alpha_0^{c_s}\epsilon_2\epsilon_3\log(c_s)^3\\
&+ 1.4588172641879\epsilon_1\epsilon_2\epsilon_3\log(c_s)\\
&+ 0.14504832807052\epsilon_2\epsilon_3\epsilon_4\log(c_s)\\
&+ 0.66666666666667\epsilon_1\epsilon_2\epsilon_3\log(c_s)^3\\
&+ 0.36481857726926\epsilon_2\epsilon_3\epsilon_4\log(c_s)^2\\
&+ 3.1889114636156\alpha_0^{c_s}\epsilon_1\epsilon_2\log(c_s)^2\\
&- 1.0\alpha_0^{c_s}\epsilon_1\epsilon_2\log(c_s)^3\\
&- 0.14527685051511\alpha_0^{c_s}\epsilon_2\epsilon_3\log(c_s)\\
&- 1.0944557318078\alpha_0^{c_s}\epsilon_2\epsilon_3\log(c_s)^2\\
&- 2.459274309077\epsilon_1\epsilon_2\epsilon_3\log(c_s)^2\\
&- 0.16666666666667\epsilon_2\epsilon_3\epsilon_4\log(c_s)^3\\
&- 0.74845687543973\alpha_0^{c_s}\epsilon_1\epsilon_2\log(c_s)
\end{aligned}
$$

### 1.2 Tensor spectrum

$$P_t(k_*) = \frac{2H_*^2}{\pi^2 M_{\rm Pl}^2}\left[\sum_{d=0}^{3} F_{t,d} + O(\mathrm{SR}^4)\right].$$

**Relative order 0.**

$$F_{t,0} = 1.0$$

**Relative order 1.**

$$F_{t,1} = -0.54072569092296\epsilon_1$$

**Relative order 2.**

$$F_{t,2} = 1.5402686460338\epsilon_1^2 - 0.25062903478191\epsilon_1\epsilon_2$$

**Relative order 3.**

$$
\begin{aligned}
F_{t,3} = {}& 1.6057835635673\epsilon_1^3 + 1.8891060194179\epsilon_1^2\epsilon_2\\
&- 0.18929040711048\epsilon_1\epsilon_2^2 - 0.18929040711048\epsilon_1\epsilon_2\epsilon_3
\end{aligned}
$$

## 2 Consistency relations

### 2.1 $n = 0$

For $p = 3$, the absolute cutoff is $W = 4$:

$$\alpha_0^t = \sum_{d=1}^{4} R_{0,d} + O(\mathrm{SR}^5).$$

**Absolute order 1.**

$$R_{0,1} = -\frac{0.125r}{c_s}$$

**Absolute order 2.**

$$
\begin{aligned}
R_{0,2} = {}& \frac{0.0078125r^2}{c_s^2} + \frac{0.125\alpha_0^s r}{c_s}\\
&+ \frac{0.28379535568268\alpha_0^{c_s} r}{c_s} - \frac{0.125\alpha_0^s r\log(c_s)}{c_s}
\end{aligned}
$$

**Absolute order 3.**

$$
\begin{aligned}
R_{0,3} = & -\frac{0.0030712246746565 r^3}{c_s^3} + \frac{0.125 \alpha_1^s r}{c_s} \\
& + \frac{0.33325502603924 \alpha_1^{c_s} r}{c_s} - \frac{0.10280837917801 \left(\alpha_0^s\right)^2 r}{c_s} \\
& - \frac{0.20825502603924 \left(\alpha_0^{c_s}\right)^2 r}{c_s} - \frac{0.041327094794504 \alpha_0^s r^2}{c_s^2} \\
& - \frac{0.061176514254839 \alpha_0^{c_s} r^2}{c_s^2} + \frac{0.015625 \alpha_0^s r^2 \log\left(c_s\right)}{c_s^2} \\
& + \frac{0.0625 \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
& + \frac{0.125 \left(\alpha_0^s\right)^2 r \log\left(c_s\right)}{c_s} \\
& - \frac{0.0625 \left(\alpha_0^s\right)^2 r \log\left(c_s\right)^2}{c_s} \\
& - \frac{0.125 \alpha_1^s r \log\left(c_s\right)}{c_s} - \frac{0.46478233218736 \alpha_0^{c_s} \alpha_0^s r}{c_s} \\
& + \frac{0.28379535568268 \alpha_0^{c_s} \alpha_0^s r \log\left(c_s\right)}{c_s}
\end{aligned}
$$

**Absolute order 4.**

$$
\begin{aligned}
R_{0,4} = & \frac{0.0014926374043055 r^4}{c_s^4} + \frac{0.125 \alpha_2^s r}{c_s} \\
& + \frac{0.076954635474829 \left(\alpha_0^{c_s}\right)^3 r}{c_s} + \frac{0.065244100954277 \left(\alpha_0^s\right)^3 r}{c_s} \\
& + \frac{0.36074999115751 \alpha_2^{c_s} r}{c_s} + \frac{0.036259602621199 \alpha_0^{c_s} r^3}{c_s^3} \\
& + \frac{0.024554893470954 \alpha_0^s r^3}{c_s^3} + \frac{0.10320649151137 \left(\alpha_0^s\right)^2 r^2}{c_s^2} \\
& + \frac{0.21414669687749 \left(\alpha_0^{c_s}\right)^2 r^2}{c_s^2} - \frac{0.11879209843874 \alpha_1^{c_s} r^2}{c_s^2} \\
& - \frac{0.060109233906372 \alpha_1^s r^2}{c_s^2} \\
& + \frac{0.015625 \left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)^2}{c_s^2} \\
& + \frac{0.0625 \alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
& + \frac{0.0625 \left(\alpha_0^s\right)^3 r \log\left(c_s\right)^2}{c_s} \\
& + \frac{0.34966216656143 \alpha_0^{c_s} \alpha_0^s r^2}{c_s^2} \\
& + \frac{0.60410986672479 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r}{c_s} \\
& + \frac{0.041327094794504 \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
& + \frac{0.46184803450948 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r}{c_s} \\
& - \frac{0.0078125 \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{0.125\alpha_2^s r \log(c_s)}{c_s} \\
&- \frac{0.0092136740239694\alpha_0^s r^3 \log(c_s)}{c_s^3} \\
&- \frac{0.10280837917801 (\alpha_0^s)^3 r \log(c_s)}{c_s} \\
&- \frac{0.51223106590429\alpha_0^{c_s}\alpha_1^s r}{c_s} - \frac{0.26465926210717\alpha_0^{c_s}\alpha_1^{c_s} r}{c_s} \\
&- \frac{0.58335484068555\alpha_0^s\alpha_1^{c_s} r}{c_s} \\
&- \frac{0.082654189589007 (\alpha_0^s)^2 r^2 \log(c_s)}{c_s^2} \\
&- \frac{0.020833333333333\alpha_2^s r \log(c_s)^3}{c_s} \\
&- \frac{0.020833333333333 (\alpha_0^s)^3 r \log(c_s)^3}{c_s} \\
&- \frac{0.23087387125098\alpha_0^s\alpha_1^s r}{c_s} \\
&+ \frac{0.0625\alpha_0^s\alpha_1^s r \log(c_s)^3}{c_s} \\
&+ \frac{0.14189767784134\alpha_0^{c_s} (\alpha_0^s)^2 r \log(c_s)^2}{c_s} \\
&+ \frac{0.46478233218736\alpha_0^{c_s}\alpha_1^s r \log(c_s)}{c_s} \\
&+ \frac{0.33061675835603\alpha_0^s\alpha_1^s r \log(c_s)}{c_s} \\
&+ \frac{0.33325502603924\alpha_0^s\alpha_1^{c_s} r \log(c_s)}{c_s} \\
&- \frac{0.1875\alpha_0^s\alpha_1^s r \log(c_s)^2}{c_s} \\
&- \frac{0.14189767784134\alpha_0^{c_s}\alpha_1^s r \log(c_s)^2}{c_s} \\
&- \frac{0.20825502603924 (\alpha_0^{c_s})^2 \alpha_0^s r \log(c_s)}{c_s} \\
&- \frac{0.46478233218736\alpha_0^{c_s} (\alpha_0^s)^2 r \log(c_s)}{c_s} \\
&- \frac{0.12235302850968\alpha_0^{c_s}\alpha_0^s r^2 \log(c_s)}{c_s^2}
\end{aligned}
$$

## 2.2 $n = 1$

For $p = 3$, the absolute cutoff is $W = 5$:

$$\alpha_1^t = \sum_{d=2}^{5} R_{1,d} + O(\mathrm{SR}^6).$$

**Absolute order 2.**

$$
\begin{aligned}
R_{1,2} = &\frac{0.015625 r^2}{c_s^2} + \frac{0.125\alpha_0^{c_s} r}{c_s} \\
&+ \frac{0.125\alpha_0^s r}{c_s}
\end{aligned}
$$

**Absolute order 3.**

$$
\begin{aligned}
R_{1,3} = & -\frac{0.0029296875 r^3}{c_s^3} + \frac{0.125 \alpha_1^s r}{c_s} \\
& + \frac{0.28379535568268 \alpha_1^{c_s} r}{c_s} - \frac{0.125 \left(\alpha_0^s\right)^2 r}{c_s} \\
& - \frac{0.046875 \alpha_0^s r^2}{c_s^2} - \frac{0.28379535568268 \left(\alpha_0^{c_s}\right)^2 r}{c_s} \\
& - \frac{0.086573838920671 \alpha_0^{c_s} r^2}{c_s^2} + \frac{0.03125 \alpha_0^s r^2 \log\left(c_s\right)}{c_s^2} \\
& + \frac{0.125 \left(\alpha_0^s\right)^2 r \log\left(c_s\right)}{c_s} \\
& - \frac{0.125 \alpha_1^s r \log\left(c_s\right)}{c_s} - \frac{0.53379535568268 \alpha_0^{c_s} \alpha_0^s r}{c_s} \\
& + \frac{0.125 \alpha_0^{c_s} \alpha_0^s r \log\left(c_s\right)}{c_s}
\end{aligned}
$$

**Absolute order 4.**

$$
\begin{aligned}
R_{1,4} = & \frac{0.0016576826498282 r^4}{c_s^4} + \frac{0.125 \alpha_2^s r}{c_s} \\
& + \frac{0.12398128438351 \left(\alpha_0^s\right)^2 r^2}{c_s^2} + \frac{0.10280837917801 \left(\alpha_0^s\right)^3 r}{c_s} \\
& + \frac{0.20825502603924 \left(\alpha_0^{c_s}\right)^3 r}{c_s} \\
& + \frac{0.25495658892655 \left(\alpha_0^{c_s}\right)^2 r^2}{c_s^2} + \frac{0.038806320518347 \alpha_0^{c_s} r^3}{c_s^3} \\
& + \frac{0.33325502603924 \alpha_2^{c_s} r}{c_s} + \frac{0.027641022071908 \alpha_0^s r^3}{c_s^3} \\
& - \frac{0.14449027076465 \alpha_1^{c_s} r^2}{c_s^2} - \frac{0.072577094794504 \alpha_1^s r^2}{c_s^2} \\
& + \frac{0.03125 \left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)^2}{c_s^2} \\
& + \frac{0.0625 \alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
& + \frac{0.0625 \left(\alpha_0^s\right)^3 r \log\left(c_s\right)^2}{c_s} \\
& + \frac{0.046875 \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
& + \frac{0.69259071136537 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r}{c_s} \\
& + \frac{0.4077766400662 \alpha_0^{c_s} \alpha_0^s r^2}{c_s^2} \\
& + \frac{0.95683271390928 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r}{c_s} \\
& - \frac{0.015625 \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
& - \frac{0.125 \alpha_2^s r \log\left(c_s\right)}{c_s} \\
& - \frac{0.125 \left(\alpha_0^s\right)^3 r \log\left(c_s\right)}{c_s}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{0.09375\left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.0087890625\alpha_0^s r^3 \log\left(c_s\right)}{c_s^3} - \frac{0.33061675835603\alpha_0^s \alpha_1^s r}{c_s} \\
&- \frac{0.71478233218736\alpha_0^{c_s} \alpha_1^s r}{c_s} - \frac{0.74976507811772\alpha_0^{c_s} \alpha_1^{c_s} r}{c_s} \\
&- \frac{0.79803735822659\alpha_0^s \alpha_1^{c_s} r}{c_s} \\
&+ \frac{0.0625\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.375\alpha_0^s \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.28379535568268\alpha_0^s \alpha_1^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.53379535568268\alpha_0^{c_s} \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.0625\alpha_0^{c_s} \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.1875\alpha_0^s \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.28379535568268\left(\alpha_0^{c_s}\right)^2 \alpha_0^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.17314767784134\alpha_0^{c_s} \alpha_0^s r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.53379535568268\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r \log\left(c_s\right)}{c_s}
\end{aligned}
$$

**Absolute order 5.**

$$
\begin{aligned}
R_{1,5} = &- \frac{0.0010528680915447 r^5}{c_s^5} + \frac{0.125\alpha_3^s r}{c_s} \\
&+ \frac{0.088617304208522\alpha_1^{c_s} r^3}{c_s^3} + \frac{0.36074999115751\alpha_3^{c_s} r}{c_s} \\
&+ \frac{0.050025543685844\alpha_1^s r^3}{c_s^3} - \frac{0.56573567099871\left(\alpha_0^{c_s}\right)^3 r^2}{c_s^2} \\
&- \frac{0.076954635474829\left(\alpha_0^{c_s}\right)^4 r}{c_s} - \frac{0.091359233906372\alpha_2^s r^2}{c_s^2} \\
&- \frac{0.065244100954277\left(\alpha_0^s\right)^4 r}{c_s} - \frac{0.20897959622812\alpha_2^{c_s} r^2}{c_s^2} \\
&- \frac{0.24604962823258\left(\alpha_0^{c_s}\right)^2 r^3}{c_s^3} \\
&- \frac{0.13027434666455\left(\alpha_0^s\right)^2 r^3}{c_s^3} \\
&- \frac{0.26465926210717\left(\alpha_1^{c_s}\right)^2 r}{c_s} - \frac{0.021075080402355\alpha_0^s r^4}{c_s^4} \\
&- \frac{0.23087387125098\left(\alpha_1^s\right)^2 r}{c_s} - \frac{0.029498514193987\alpha_0^{c_s} r^4}{c_s^4} \\
&- \frac{0.24842610305581\left(\alpha_0^s\right)^3 r^2}{c_s^2} \\
&+ \frac{0.0625\alpha_3^s r \log\left(c_s\right)^2}{c_s}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{0.0625 \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.00439453125 \alpha_1^s r^3 \log\left(c_s\right)^2}{c_s^3} \\
&+ \frac{1.5811232648164 \left(\alpha_0^{c_s}\right)^2 \alpha_1^s r}{c_s} \\
&+ \frac{0.24796256876702 \left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.082923066215725 \left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{0.71021433456542 \alpha_0^{c_s} \alpha_1^s r^2}{c_s^2} \\
&+ \frac{0.10280837917801 \left(\alpha_0^s\right)^4 r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.42660617411381 \left(\alpha_0^s\right)^2 \alpha_1^s r}{c_s} \\
&+ \frac{1.045202875195 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r}{c_s} \\
&+ \frac{0.072577094794504 \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.33061675835603 \left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.0052083333333333 \alpha_2^s r^2 \log\left(c_s\right)^3}{c_s^2} \\
&+ \frac{0.020833333333333 \left(\alpha_0^s\right)^4 r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.020833333333333 \left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)^3}{c_s^2} \\
&+ \frac{0.81639883012011 \alpha_0^s \alpha_1^{c_s} r^2}{c_s^2} \\
&+ \frac{0.41559991864823 \alpha_0^s \alpha_1^s r^2}{c_s^2} \\
&+ \frac{0.0066307305993129 \alpha_0^s r^4 \log\left(c_s\right)}{c_s^4} \\
&+ \frac{0.92119486345495 \alpha_0^{c_s} \alpha_1^{c_s} r^2}{c_s^2} \\
&+ \frac{0.49552316853166 \left(\alpha_0^{c_s}\right)^2 \alpha_1^{c_s} r}{c_s} \\
&- \frac{0.0625 \left(\alpha_0^s\right)^4 r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.125 \alpha_3^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.0234375 \alpha_2^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&- \frac{0.09375 \left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)^2}{c_s^2} \\
&- \frac{0.1875 \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^2}{c_s}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{0.01318359375\left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)^2}{c_s^3} \\
&- \frac{0.62990051464178\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r}{c_s} \\
&- \frac{0.62540925326469\alpha_0^{c_s}\alpha_2^{c_s} r}{c_s} \\
&- \frac{0.39180031621678\alpha_0^{c_s}\alpha_0^s r^3}{c_s^3} - \frac{1.0955859065898\alpha_1^{c_s}\alpha_1^s r}{c_s} \\
&- \frac{1.5307402334216\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r}{c_s} \\
&- \frac{1.7168681131348\left(\alpha_0^{c_s}\right)^2\alpha_0^s r^2}{c_s^2} \\
&- \frac{0.88931952823886\left(\alpha_0^{c_s}\right)^3\alpha_0^s r}{c_s} \\
&- \frac{1.2784021784808\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^2}{c_s^2} \\
&- \frac{0.020833333333333\alpha_3^s r\log\left(c_s\right)^3}{c_s} - \frac{0.35587387125098\alpha_0^s\alpha_2^s r}{c_s} \\
&- \frac{0.76223106590429\alpha_0^{c_s}\alpha_2^s r}{c_s} \\
&- \frac{0.027641022071908\alpha_1^s r^3\log\left(c_s\right)}{c_s^3} \\
&- \frac{0.94410483184306\alpha_0^s\alpha_2^{c_s} r}{c_s} \\
&+ \frac{0.375\left(\alpha_0^s\right)^2\alpha_1^s r\log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.140625\alpha_0^s\alpha_1^s r^2\log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.14189767784134\left(\alpha_0^s\right)^2\alpha_1^{c_s} r\log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.14189767784134\left(\alpha_0^{c_s}\right)^2\alpha_1^s r\log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.086573838920671\alpha_0^{c_s}\alpha_1^s r^2\log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.69259071136537\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r\log\left(c_s\right)}{c_s} \\
&+ \frac{0.20825502603924\left(\alpha_0^{c_s}\right)^3\alpha_0^s r\log\left(c_s\right)}{c_s} \\
&+ \frac{0.11641896155504\alpha_0^{c_s}\alpha_0^s r^3\log\left(c_s\right)}{c_s^3} \\
&+ \frac{1.9974177645303\alpha_0^{c_s}\alpha_0^s\alpha_1^s r}{c_s} \\
&+ \frac{0.5099131778531\left(\alpha_0^{c_s}\right)^2\alpha_0^s r^2\log\left(c_s\right)}{c_s^2} \\
&+ \frac{2.3894888622815\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s} r}{c_s} \\
&+ \frac{0.33325502603924\alpha_0^s\alpha_2^{c_s} r\log\left(c_s\right)}{c_s} \\
&+ \frac{0.020833333333333\alpha_0^{c_s}\alpha_2^s r\log\left(c_s\right)^3}{c_s}
\end{aligned}
$$

$$+\frac{0.020833333333333\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.083333333333333\alpha_0^s\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.71478233218736\alpha_0^{c_s}\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.79803735822659\alpha_1^{c_s}\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.81555328013239\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.45561675835603\alpha_0^s\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.95683271390928\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r\log\left(c_s\right)}{c_s}$$
$$-\frac{0.03125\alpha_0^s\alpha_1^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$-\frac{0.125\left(\alpha_0^s\right)^2\alpha_1^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.25\alpha_0^s\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.14189767784134\alpha_1^{c_s}\alpha_1^s r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.14189767784134\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.17314767784134\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^2\log\left(c_s\right)^2}{c_s^2}$$
$$-\frac{0.26689767784134\alpha_0^{c_s}\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.26689767784134\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.2889805415293\alpha_0^s\alpha_1^{c_s} r^2\log\left(c_s\right)}{c_s^2}$$
$$-\frac{0.63904189589007\left(\alpha_0^s\right)^2\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$-\frac{0.39311675835603\alpha_0^s\alpha_1^s r^2\log\left(c_s\right)}{c_s^2}$$
$$-\frac{0.79803735822659\left(\alpha_0^s\right)^2\alpha_1^{c_s} r\log\left(c_s\right)}{c_s}$$
$$-\frac{0.4077766400662\alpha_0^{c_s}\alpha_1^s r^2\log\left(c_s\right)}{c_s^2}$$

$$-\frac{0.95683271390928\left(\alpha_0^{c_s}\right)^2\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.80069303352403\alpha_0^{c_s}\alpha_0^s\alpha_1^s r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.0625\alpha_0^{c_s}\alpha_0^s\alpha_1^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{2.0999637549181\alpha_0^{c_s}\alpha_0^s\alpha_1^s r\log\left(c_s\right)}{c_s}$$

$$- \frac{0.74976507811772\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s} r\log(c_s)}{c_s}$$

## 2.3 $n = 2$

For $p = 3$, the absolute cutoff is $W = 6$:

$$\alpha_2^t = \sum_{d=3}^{6} R_{2,d} + O(\mathrm{SR}^7).$$

**Absolute order 3.**

$$\begin{aligned} R_{2,3} = &- \frac{0.00390625r^3}{c_s^3} + \frac{0.125\alpha_1^{c_s} r}{c_s} \\ &+ \frac{0.125\alpha_1^s r}{c_s} - \frac{0.125\left(\alpha_0^{c_s}\right)^2 r}{c_s} \\ &- \frac{0.125\left(\alpha_0^s\right)^2 r}{c_s} - \frac{0.046875\alpha_0^{c_s} r^2}{c_s^2} \\ &- \frac{0.046875\alpha_0^s r^2}{c_s^2} - \frac{0.25\alpha_0^{c_s}\alpha_0^s r}{c_s} \end{aligned}$$

**Absolute order 4.**

$$\begin{aligned} R_{2,4} = &\frac{0.0013427734375r^4}{c_s^4} + \frac{0.125\alpha_2^s r}{c_s} \\ &+ \frac{0.125\left(\alpha_0^s\right)^3 r}{c_s} + \frac{0.125\left(\alpha_0^s\right)^2 r^2}{c_s^2} \\ &+ \frac{0.025390625\alpha_0^s r^3}{c_s^3} + \frac{0.28379535568268\alpha_2^{c_s} r}{c_s} \\ &+ \frac{0.28379535568268\left(\alpha_0^{c_s}\right)^3 r}{c_s} \\ &+ \frac{0.24409651676201\left(\alpha_0^{c_s}\right)^2 r^2}{c_s^2} + \frac{0.040277689595252\alpha_0^{c_s} r^3}{c_s^3} \\ &- \frac{0.0625\alpha_1^s r^2}{c_s^2} - \frac{0.12204825838101\alpha_1^{c_s} r^2}{c_s^2} \\ &+ \frac{0.046875\alpha_1^s r^2\log(c_s)}{c_s^2} \\ &+ \frac{0.94259071136537\left(\alpha_0^{c_s}\right)^2\alpha_0^s r}{c_s} \\ &+ \frac{0.78379535568268\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r}{c_s} \\ &+ \frac{0.41597151676201\alpha_0^{c_s}\alpha_0^s r^2}{c_s^2} - \frac{0.125\alpha_2^s r\log(c_s)}{c_s} \\ &- \frac{0.125\left(\alpha_0^s\right)^3 r\log(c_s)}{c_s} \\ &- \frac{0.01171875\alpha_0^s r^3\log(c_s)}{c_s^3} \\ &- \frac{0.09375\left(\alpha_0^s\right)^2 r^2\log(c_s)}{c_s^2} - \frac{0.375\alpha_0^s\alpha_1^s r}{c_s} \\ &- \frac{0.81759071136537\alpha_0^s\alpha_1^{c_s} r}{c_s} - \frac{0.78379535568268\alpha_0^{c_s}\alpha_1^s r}{c_s} \\ &- \frac{0.85138606704805\alpha_0^{c_s}\alpha_1^{c_s} r}{c_s} \end{aligned}$$

$$
\begin{aligned}
&+\frac{0.125\alpha_0^s\alpha_1^{c_s}r\log\left(c_s\right)}{c_s}\\
&+\frac{0.25\alpha_0^{c_s}\alpha_1^s r\log\left(c_s\right)}{c_s}\\
&+\frac{0.375\alpha_0^s\alpha_1^s r\log\left(c_s\right)}{c_s}\\
&-\frac{0.125\left(\alpha_0^{c_s}\right)^2\alpha_0^s r\log\left(c_s\right)}{c_s}\\
&-\frac{0.25\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r\log\left(c_s\right)}{c_s}\\
&-\frac{0.09375\alpha_0^{c_s}\alpha_0^s r^2\log\left(c_s\right)}{c_s^2}
\end{aligned}
$$

**Absolute order 5.**

$$
\begin{aligned}
R_{2,5} = &-\frac{0.00099348164677838r^5}{c_s^5}+\frac{0.125\alpha_3^s r}{c_s}\\
&+\frac{0.050668108270534\alpha_1^s r^3}{c_s^3}+\frac{0.33325502603924\alpha_3^{c_s}r}{c_s}\\
&+\frac{0.087560258989506\alpha_1^{c_s}r^3}{c_s^3}-\frac{0.24566880066715\left(\alpha_0^{c_s}\right)^2 r^3}{c_s^3}\\
&-\frac{0.13499234851023\left(\alpha_0^s\right)^2 r^3}{c_s^3}\\
&-\frac{0.28928966356152\left(\alpha_0^s\right)^3 r^2}{c_s^2}-\frac{0.020502543360439\alpha_0^s r^4}{c_s^4}\\
&-\frac{0.10280837917801\left(\alpha_0^s\right)^4 r}{c_s}-\frac{0.20825502603924\left(\alpha_0^{c_s}\right)^4 r}{c_s}\\
&-\frac{0.027325781299929\alpha_0^{c_s}r^4}{c_s^4}-\frac{0.64251673826997\left(\alpha_0^{c_s}\right)^3 r^2}{c_s^2}\\
&-\frac{0.33061675835603\left(\alpha_1^s\right)^2 r}{c_s}-\frac{0.088202094794504\alpha_2^s r^2}{c_s^2}\\
&-\frac{0.18614714901955\alpha_2^{c_s}r^2}{c_s^2}-\frac{0.74976507811772\left(\alpha_1^{c_s}\right)^2 r}{c_s}\\
&+\frac{0.0625\alpha_2^s r^2\log\left(c_s\right)}{c_s^2}+\frac{0.0625\alpha_3^s r\log\left(c_s\right)^2}{c_s}\\
&+\frac{0.125\left(\alpha_0^s\right)^4 r\log\left(c_s\right)}{c_s}\\
&+\frac{0.25\left(\alpha_0^s\right)^3 r^2\log\left(c_s\right)}{c_s^2}\\
&+\frac{0.005859375\alpha_1^s r^3\log\left(c_s\right)^2}{c_s^3}\\
&+\frac{0.375\left(\alpha_1^s\right)^2 r\log\left(c_s\right)}{c_s}\\
&+\frac{0.00537109375\alpha_0^s r^4\log\left(c_s\right)}{c_s^4}\\
&+\frac{0.076171875\left(\alpha_0^s\right)^2 r^3\log\left(c_s\right)}{c_s^3}\\
&+\frac{2.2054104017793\left(\alpha_0^{c_s}\right)^2\alpha_1^s r}{c_s}\\
&+\frac{1.490628069592\left(\alpha_0^s\right)^2\alpha_1^{c_s}r}{c_s}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{1.0148110363091 \alpha_0^{c_s} \alpha_1^{c_s} r^2}{c_s^2} \\
&+ \frac{0.63904189589007 \left(\alpha_0^s\right)^2 \alpha_1^s r}{c_s} \\
&+ \frac{1.3745301562354 \left(\alpha_0^{c_s}\right)^2 \alpha_1^{c_s} r}{c_s} \\
&+ \frac{0.74025304063896 \alpha_0^{c_s} \alpha_1^s r^2}{c_s^2} \\
&+ \frac{0.87364314908906 \alpha_0^s \alpha_1^{c_s} r^2}{c_s^2} \\
&+ \frac{0.46569385315053 \alpha_0^s \alpha_1^s r^2}{c_s^2} \\
&- \frac{0.0625 \left(\alpha_0^s\right)^4 r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.125 \alpha_3^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.0234375 \alpha_2^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&- \frac{0.09375 \left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)^2}{c_s^2} \\
&- \frac{0.1875 \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.017578125 \left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)^2}{c_s^3} \\
&- \frac{0.025390625 \alpha_1^s r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{0.92039909054338 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 r}{c_s} \\
&- \frac{1.1312923842658 \alpha_0^s \alpha_2^{c_s} r}{c_s} \\
&- \frac{1.4488830956312 \left(\alpha_0^{c_s}\right)^3 \alpha_0^s r}{c_s} \\
&- \frac{2.1832187809573 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r}{c_s} \\
&- \frac{1.083020104157 \alpha_0^{c_s} \alpha_2^{c_s} r}{c_s} \\
&- \frac{1.4169873656614 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r^2}{c_s^2} \\
&- \frac{1.512819690414 \alpha_1^{c_s} \alpha_1^s r}{c_s} - \frac{0.38983219984103 \alpha_0^{c_s} \alpha_0^s r^3}{c_s^3} \\
&- \frac{0.45561675835603 \alpha_0^s \alpha_2^s r}{c_s} \\
&- \frac{1.8893109571319 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^2}{c_s^2} \\
&- \frac{0.96478233218736 \alpha_0^{c_s} \alpha_2^s r}{c_s} \\
&+ \frac{0.0625 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.0625 \left(\alpha_0^{c_s}\right)^2 \alpha_1^s r \log\left(c_s\right)^2}{c_s}
\end{aligned}
$$

$$+\frac{0.5\alpha_0^s\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.046875\alpha_0^{c_s}\alpha_1^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.375\left(\alpha_0^s\right)^2\alpha_1^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.140625\alpha_0^s\alpha_1^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.28379535568268\alpha_0^s\alpha_2^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.28379535568268\left(\alpha_0^{c_s}\right)^3\alpha_0^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.81759071136537\alpha_1^{c_s}\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.94259071136537\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.78379535568268\alpha_0^{c_s}\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.78379535568268\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.48819303352403\left(\alpha_0^{c_s}\right)^2\alpha_0^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.83194303352403\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.12083306878576\alpha_0^{c_s}\alpha_0^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{2.8055805132741\alpha_0^{c_s}\alpha_0^s\alpha_1^s r}{c_s}$$
$$+\frac{3.7452632198456\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s} r}{c_s}$$
$$-\frac{0.0625\alpha_1^{c_s}\alpha_1^s r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.0625\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.125\alpha_0^{c_s}\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.125\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.25\alpha_0^s\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.09375\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^2\log\left(c_s\right)^2}{c_s^2}$$
$$-\frac{0.375\alpha_0^s\alpha_1^s r^2\log\left(c_s\right)}{c_s^2}$$
$$-\frac{0.75\left(\alpha_0^s\right)^2\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$-\frac{0.81759071136537\left(\alpha_0^s\right)^2\alpha_1^{c_s} r\log\left(c_s\right)}{c_s}$$
$$-\frac{0.94259071136537\left(\alpha_0^{c_s}\right)^2\alpha_1^s r\log\left(c_s\right)}{c_s}$$

$$
\begin{aligned}
&- \frac{0.24409651676201\alpha_0^s\alpha_1^{c_s}r^2\log(c_s)}{c_s^2} \\
&- \frac{0.41597151676201\alpha_0^{c_s}\alpha_1^s r^2\log(c_s)}{c_s^2} \\
&+ \frac{0.375\alpha_0^{c_s}\alpha_0^s\alpha_1^s r\log(c_s)^2}{c_s} \\
&- \frac{0.85138606704805\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s}r\log(c_s)}{c_s} \\
&- \frac{2.3513860670481\alpha_0^{c_s}\alpha_0^s\alpha_1^s r\log(c_s)}{c_s}
\end{aligned}
$$

**Absolute order 6.**

$$
\begin{aligned}
R_{2,6} = &\frac{0.0007834832445242r^6}{c_s^6} + \frac{0.125\alpha_4^s r}{c_s} \\
&+ \frac{0.076954635474829\left(\alpha_0^{c_s}\right)^5 r}{c_s} + \frac{0.065244100954277\left(\alpha_0^s\right)^5 r}{c_s} \\
&+ \frac{0.51447193930681\left(\alpha_0^s\right)^3 r^3}{c_s^3} + \frac{0.018883632267472\alpha_0^s r^5}{c_s^5} \\
&+ \frac{0.36074999115751\alpha_4^{c_s} r}{c_s} + \frac{1.0488534998662\left(\alpha_1^{c_s}\right)^2 r^2}{c_s^2} \\
&+ \frac{0.15473919538016\alpha_2^{c_s} r^3}{c_s^3} + \frac{0.077748164662437\alpha_2^s r^3}{c_s^3} \\
&+ \frac{1.2689136192411\left(\alpha_0^{c_s}\right)^4 r^2}{c_s^2} + \frac{0.025194043126048\alpha_0^{c_s} r^5}{c_s^5} \\
&+ \frac{0.5388653261447\left(\alpha_0^s\right)^4 r^2}{c_s^2} \\
&+ \frac{0.27180459639332\left(\alpha_0^{c_s}\right)^2 r^4}{c_s^4} \\
&+ \frac{1.1085144407626\left(\alpha_0^{c_s}\right)^3 r^3}{c_s^3} \\
&+ \frac{0.15691226448912\left(\alpha_0^s\right)^2 r^4}{c_s^4} \\
&+ \frac{0.4600841525546\left(\alpha_1^s\right)^2 r^2}{c_s^2} - \frac{0.10698423390637\alpha_3^s r^2}{c_s^2} \\
&- \frac{0.044329625847117\alpha_1^s r^4}{c_s^4} - \frac{0.072500515380315\alpha_1^{c_s} r^4}{c_s^4} \\
&- \frac{0.25407334512281\alpha_3^{c_s} r^2}{c_s^2} + \frac{0.0078125\alpha_3^s r^2\log(c_s)^3}{c_s^2} \\
&+ \frac{0.0625\alpha_4^s r\log(c_s)^2}{c_s} \\
&+ \frac{0.0625\left(\alpha_0^s\right)^5 r\log(c_s)^2}{c_s} \\
&+ \frac{0.25\left(\alpha_0^s\right)^4 r^2\log(c_s)^2}{c_s^2} \\
&+ \frac{0.1875\left(\alpha_1^s\right)^2 r^2\log(c_s)^2}{c_s^2} \\
&+ \frac{0.0107421875\left(\alpha_0^s\right)^2 r^4\log(c_s)^2}{c_s^4} \\
&+ \frac{0.0126953125\alpha_2^s r^3\log(c_s)^2}{c_s^3}
\end{aligned}
$$

$$+\frac{0.1142578125\left(\alpha_0^s\right)^3 r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{1.4391217537063\alpha_0^s\alpha_2^{c_s} r^2}{c_s^2}$$
$$+\frac{5.812063941528\left(\alpha_0^{c_s}\right)^3 \alpha_0^s r^2}{c_s^2}$$
$$+\frac{7.7501463158245\left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r^2}{c_s^2}$$
$$+\frac{0.78248004536479\left(\alpha_0^s\right)^2 \alpha_2^s r}{c_s}$$
$$+\frac{1.111888199871\alpha_0^{c_s}\alpha_2^s r^2}{c_s^2}$$
$$+\frac{1.1745291897529\left(\alpha_0^{c_s}\right)^4 \alpha_0^s r}{c_s}$$
$$+\frac{1.2557055991705\alpha_0^{c_s}\left(\alpha_1^{c_s}\right)^2 r}{c_s}$$
$$+\frac{1.7406927007245\alpha_1^{c_s}\alpha_1^s r^2}{c_s^2}$$
$$+\frac{1.1209324217963\left(\alpha_0^{c_s}\right)^2 \alpha_2^{c_s} r}{c_s}$$
$$+\frac{0.020502543360439\alpha_1^s r^4 \log\left(c_s\right)}{c_s^4}$$
$$+\frac{0.79795299477407\alpha_0^{c_s}\left(\alpha_0^s\right)^4 r}{c_s}$$
$$+\frac{3.6749700194 14\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r^2}{c_s^2}$$
$$+\frac{1.0840862194786\alpha_0^s\left(\alpha_1^s\right)^2 r}{c_s}$$
$$+\frac{2.3085962344727\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^3}{c_s^3}$$
$$+\frac{2.5589083941373\alpha_0^{c_s}\left(\alpha_1^s\right)^2 r}{c_s}$$
$$+\frac{2.6541481243887\alpha_0^s\left(\alpha_1^{c_s}\right)^2 r}{c_s}$$
$$+\frac{0.67405262036735\alpha_0^s\alpha_2^s r^2}{c_s^2}$$
$$+\frac{0.088202094794504\alpha_3^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$+\frac{2.8532314594288\left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^3 r}{c_s}$$
$$+\frac{2.995589067438\left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^3}{c_s^3}$$
$$+\frac{3.0581366629081\left(\alpha_0^{c_s}\right)^2 \alpha_2^s r}{c_s}$$
$$+\frac{1.5570001901118\alpha_0^{c_s}\alpha_2^{c_s} r^2}{c_s^2}$$
$$+\frac{3.3768924755698\left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^2 r}{c_s}$$

$$+\frac{0.43745143869001\alpha_0^{c_s}\alpha_0^s r^4}{c_s^4}$$
$$+\frac{1.9893077070381\left(\alpha_0^s\right)^2\alpha_2^{c_s} r}{c_s}$$
$$-\frac{0.001953125\alpha_2^s r^3\log\left(c_s\right)^3}{c_s^3}$$
$$-\frac{0.03125\alpha_3^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$-\frac{0.0625\left(\alpha_0^s\right)^4 r^2\log\left(c_s\right)^3}{c_s^2}$$
$$-\frac{0.125\alpha_4^s r\log\left(c_s\right)}{c_s}$$
$$-\frac{0.046875\left(\alpha_1^s\right)^2 r^2\log\left(c_s\right)^3}{c_s^2}$$
$$-\frac{0.017578125\left(\alpha_0^s\right)^3 r^3\log\left(c_s\right)^3}{c_s^3}$$
$$-\frac{0.002685546875\alpha_1^s r^4\log\left(c_s\right)^2}{c_s^4}$$
$$-\frac{0.98015911825867\alpha_0^s\alpha_1^{c_s} r^3}{c_s^3}$$
$$-\frac{4.0010774285941\left(\alpha_0^{c_s}\right)^2\alpha_1^{c_s} r^2}{c_s^2}$$
$$-\frac{4.071100917586\left(\alpha_0^{c_s}\right)^2\alpha_1^s r^2}{c_s^2}$$
$$-\frac{0.0049674082338919\alpha_0^s r^5\log\left(c_s\right)}{c_s^5}$$
$$-\frac{3.2853576417753\left(\alpha_0^s\right)^2\alpha_1^{c_s} r^2}{c_s^2}$$
$$-\frac{0.84666127388087\alpha_0^{c_s}\alpha_1^s r^3}{c_s^3}-\frac{2.0396907384329\alpha_1^s\alpha_2^{c_s} r}{c_s}$$
$$-\frac{0.56686562073618\alpha_0^s\alpha_1^s r^3}{c_s^3}$$
$$-\frac{0.57857932712305\left(\alpha_0^s\right)^4 r^2\log\left(c_s\right)}{c_s^2}$$
$$-\frac{1.1743023189517\alpha_0^{c_s}\alpha_1^{c_s} r^3}{c_s^3}$$
$$-\frac{0.082010173441755\left(\alpha_0^s\right)^2 r^4\log\left(c_s\right)}{c_s^4}$$
$$-\frac{0.40497704553068\left(\alpha_0^s\right)^3 r^3\log\left(c_s\right)}{c_s^3}$$
$$-\frac{0.050668108270534\alpha_2^s r^3\log\left(c_s\right)}{c_s^3}$$
$$-\frac{0.10280837917801\left(\alpha_0^s\right)^5 r\log\left(c_s\right)}{c_s}$$
$$-\frac{3.4272755069646\left(\alpha_0^{c_s}\right)^3\alpha_1^s r}{c_s}$$
$$-\frac{1.0122310659043\alpha_0^{c_s}\alpha_3^s r}{c_s}-\frac{1.154727777479\alpha_1^{c_s}\alpha_2^{c_s} r}{c_s}$$

$$
\begin{aligned}
&- \frac{1.3048548230006\alpha_0^s \alpha_3^{c_s} r}{c_s} \\
&- \frac{0.020833333333333\alpha_4^s r \log\left(c_s\right)^3}{c_s} \\
&- \frac{0.020833333333333 \left(\alpha_0^s\right)^5 r \log\left(c_s\right)^3}{c_s} \\
&- \frac{0.68758257793092 \left(\alpha_0^s\right)^3 \alpha_1^s r}{c_s} \\
&- \frac{0.80334171043098 \left(\alpha_0^{c_s}\right)^3 \alpha_1^{c_s} r}{c_s} \\
&- \frac{0.81762161375293\alpha_1^s \alpha_2^s r}{c_s} \\
&- \frac{1.6751033898368 \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r}{c_s} \\
&- \frac{1.7284662943263 \left(\alpha_0^s\right)^2 \alpha_1^s r^2}{c_s^2} \\
&- \frac{1.8578169724941\alpha_1^{c_s} \alpha_2^s r}{c_s} \\
&- \frac{0.46569385315053 \left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.48087387125098\alpha_0^s \alpha_3^s r}{c_s} - \frac{0.9861592444222\alpha_0^{c_s} \alpha_3^{c_s} r}{c_s} \\
&+ \frac{0.25\alpha_0^s \alpha_2^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.1875 \left(\alpha_0^s\right)^2 \alpha_1^s r^2 \log\left(c_s\right)^3}{c_s^2} \\
&+ \frac{0.625 \left(\alpha_0^s\right)^2 \alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.017578125\alpha_0^s \alpha_1^s r^3 \log\left(c_s\right)^3}{c_s^3} \\
&+ \frac{0.9375\alpha_0^s \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.14189767784134 \left(\alpha_0^s\right)^2 \alpha_2^{c_s} r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.14189767784134 \left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^2 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.47129535568268 \left(\alpha_0^{c_s}\right)^2 \alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.47129535568268 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^3 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.39189767784134\alpha_0^{c_s} \left(\alpha_0^s\right)^4 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.12204825838101\alpha_1^{c_s} \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.48819303352403 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.20798575838101\alpha_0^{c_s} \alpha_2^s r^2 \log\left(c_s\right)^2}{c_s^2}
\end{aligned}
$$

$$+\frac{0.83194303352403\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{1.1312923842658\alpha_1^s\alpha_2^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.175693033524\alpha_0^{c_s}\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{1.4488830956312\left(\alpha_0^{c_s}\right)^3\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.490628069592\left(\alpha_0^s\right)^3\alpha_1^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.26268077696852\alpha_0^s\alpha_1^{c_s} r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{0.18124960317863\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{1.1168502750681\alpha_1^s\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.42198902183206\alpha_0^s\alpha_1^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{1.0502754126021\left(\alpha_0^s\right)^3\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{4.2922579734285\alpha_0^{c_s}\alpha_0^s\alpha_2^{c_s} r}{c_s}$$
$$+\frac{0.58061675835603\alpha_0^s\alpha_3^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{5.1834094215102\alpha_0^s\alpha_1^{c_s}\alpha_1^s r}{c_s}$$
$$+\frac{0.33325502603924\alpha_0^s\alpha_3^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.020833333333333\alpha_1^{c_s}\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.020833333333333\left(\alpha_0^s\right)^3\alpha_1^{c_s} r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.041666666666667\alpha_0^{c_s}\alpha_3^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{1.512819690414\alpha_1^{c_s}\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.38983219984103\alpha_0^{c_s}\alpha_1^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{0.10416666666667\alpha_0^s\alpha_3^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.20833333333333\alpha_1^s\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.20833333333333\left(\alpha_0^s\right)^3\alpha_1^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{1.7472862981781\left(\alpha_0^s\right)^2\alpha_1^{c_s} r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{3.5711394600416\alpha_0^{c_s}\alpha_0^s\alpha_2^s r}{c_s}$$
$$+\frac{1.7992566969856\left(\alpha_0^s\right)^2\alpha_1^s r^2\log\left(c_s\right)}{c_s^2}$$

$$+\frac{7.4453586567308\alpha_0^{c_s}\alpha_1^{c_s}\alpha_1^s r}{c_s}$$
$$+\frac{1.8893109571319\left(\alpha_0^{c_s}\right)^2\alpha_1^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.96478233218736\alpha_0^{c_s}\alpha_3^s r\log\left(c_s\right)}{c_s}$$
$$-\frac{0.015625\alpha_0^{c_s}\alpha_2^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$-\frac{0.0625\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r^2\log\left(c_s\right)^3}{c_s^2}$$
$$-\frac{0.0625\alpha_0^s\alpha_2^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$-\frac{0.125\alpha_0^{c_s}\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.75\left(\alpha_0^s\right)^2\alpha_1^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$-\frac{0.3125\alpha_0^s\alpha_3^s r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.3125\alpha_0^s\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.625\alpha_1^s\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.625\left(\alpha_0^s\right)^3\alpha_1^s r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.1142578125\alpha_0^s\alpha_1^s r^3\log\left(c_s\right)^2}{c_s^3}$$
$$-\frac{2.2054104017793\left(\alpha_0^{c_s}\right)^2\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$-\frac{0.14189767784134\alpha_1^s\alpha_2^{c_s} r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.14189767784134\left(\alpha_0^{c_s}\right)^3\alpha_1^s r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.40879535568268\alpha_1^{c_s}\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.40879535568268\left(\alpha_0^s\right)^3\alpha_1^{c_s} r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.39189767784134\alpha_0^{c_s}\alpha_3^s r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.24409651676201\left(\alpha_0^s\right)^2\alpha_1^{c_s} r^2\log\left(c_s\right)^2}{c_s^2}$$
$$-\frac{0.24409651676201\left(\alpha_0^{c_s}\right)^2\alpha_1^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$-\frac{0.92039909054338\alpha_0^{c_s}\left(\alpha_0^s\right)^4 r\log\left(c_s\right)}{c_s}$$
$$-\frac{1.1312923842658\left(\alpha_0^s\right)^2\alpha_2^{c_s} r\log\left(c_s\right)}{c_s}$$
$$-\frac{1.4488830956312\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^2 r\log\left(c_s\right)}{c_s}$$

$$- \frac{0.060416534392878 \alpha_0^{c_s} \alpha_1^s r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{2.1832187809573 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^3 r \log\left(c_s\right)}{c_s}$$
$$- \frac{2.8055805132741 \alpha_0^{c_s} \left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s}$$
$$- \frac{4.9527673784595 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^s r}{c_s}$$
$$- \frac{1.6087005501362 \alpha_0^s \left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s}$$
$$- \frac{8.2325850087098 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} r^2}{c_s^2}$$
$$- \frac{1.0946586542461 \left(\alpha_0^s\right)^2 \alpha_2^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.64209804273954 \alpha_0^s \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{0.73700640200144 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{6.3027356936475 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^{c_s} r}{c_s}$$
$$- \frac{0.20825502603924 \left(\alpha_0^{c_s}\right)^4 \alpha_0^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.10930312519972 \alpha_0^{c_s} \alpha_0^s r^4 \log\left(c_s\right)}{c_s^4}$$
$$- \frac{8.7399852511081 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^s r}{c_s}$$
$$- \frac{1.1694965995231 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)}{c_s^3}$$
$$- \frac{1.2850334765399 \left(\alpha_0^{c_s}\right)^3 \alpha_0^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{0.020833333333333 \left(\alpha_0^{c_s}\right)^2 \alpha_2^s r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{0.020833333333333 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^3 r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{0.041666666666667 \alpha_0^{c_s} \left(\alpha_0^s\right)^4 r \log\left(c_s\right)^3}{c_s}$$
$$- \frac{2.8339747313229 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{5.8521124624951 \alpha_0^{c_s} \alpha_0^s \alpha_1^s r^2}{c_s^2}$$

$$- \frac{0.74025304063896 \alpha_0^{c_s} \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{0.37229429803911 \alpha_0^s \alpha_2^{c_s} r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{0.74976507811772 \alpha_0^s \left(\alpha_1^{c_s}\right)^2 r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.20833333333333 \left(\alpha_0^s\right)^2 \alpha_2^s r \log\left(c_s\right)^3}{c_s}$$

$$
\begin{aligned}
&- \frac{0.87364314908906\alpha_1^{c_s}\alpha_1^s r^2 \log(c_s)}{c_s^2} \\
&- \frac{7.2942095625464\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^{c_s} r}{c_s} \\
&- \frac{3.7786219142638\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r^2 \log(c_s)}{c_s^2} \\
&+ \frac{0.0625\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_1^s r \log(c_s)^3}{c_s} \\
&+ \frac{0.25\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^s r \log(c_s)^3}{c_s} \\
&+ \frac{0.09375\alpha_0^{c_s}\alpha_0^s\alpha_1^s r^2 \log(c_s)^3}{c_s^2} \\
&+ \frac{1.5675907113654\alpha_0^{c_s}\alpha_0^s\alpha_2^s r \log(c_s)^2}{c_s} \\
&+ \frac{0.42569303352403\alpha_0^{c_s}\alpha_1^{c_s}\alpha_1^s r \log(c_s)^2}{c_s} \\
&+ \frac{1.2263860670481\alpha_0^s\alpha_1^{c_s}\alpha_1^s r \log(c_s)^2}{c_s} \\
&+ \frac{5.5667777849043\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^s r \log(c_s)}{c_s} \\
&+ \frac{6.571847963694\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_1^s r \log(c_s)}{c_s} \\
&+ \frac{3.7452632198456\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^{c_s} r \log(c_s)}{c_s} \\
&+ \frac{4.3144808126008\alpha_0^{c_s}\alpha_0^s\alpha_1^s r^2 \log(c_s)}{c_s^2} \\
&+ \frac{2.0296220726182\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s} r^2 \log(c_s)}{c_s^2} \\
&+ \frac{1.3745301562354\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_1^{c_s} r \log(c_s)}{c_s} \\
&- \frac{0.0625\alpha_0^s\alpha_1^{c_s}\alpha_1^s r \log(c_s)^3}{c_s} \\
&- \frac{3.7703628454615\alpha_0^{c_s}\alpha_0^s\alpha_2^s r \log(c_s)}{c_s} \\
&- \frac{0.42569303352403\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^{c_s} r \log(c_s)^2}{c_s} \\
&- \frac{1.4138860670481\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_1^s r \log(c_s)^2}{c_s} \\
&- \frac{2.3513860670481\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^s r \log(c_s)^2}{c_s} \\
&- \frac{4.4940758295979\alpha_0^s\alpha_1^{c_s}\alpha_1^s r \log(c_s)}{c_s} \\
&- \frac{1.247914550286\alpha_0^{c_s}\alpha_0^s\alpha_1^s r^2 \log(c_s)^2}{c_s^2} \\
&- \frac{3.7452632198456\alpha_0^{c_s}\alpha_1^{c_s}\alpha_1^s r \log(c_s)}{c_s} \\
&- \frac{1.083020104157\alpha_0^{c_s}\alpha_0^s\alpha_2^{c_s} r \log(c_s)}{c_s} \\
&- \frac{0.16666666666667\alpha_0^{c_s}\alpha_0^s\alpha_2^s r \log(c_s)^3}{c_s}
\end{aligned}
$$

## 2.4 $n = 3$

For $p = 3$, the absolute cutoff is $W = 7$:

$$\alpha_3^t = \sum_{d=4}^{7} R_{3,d} + O(\mathrm{SR}^8).$$

**Absolute order 4.**

$$\begin{aligned}
R_{3,4} = {} & \frac{0.00146484375 r^4}{c_s^4} + \frac{0.125 \alpha_2^{c_s} r}{c_s} \\
& + \frac{0.125 \alpha_2^s r}{c_s} + \frac{0.125 \left(\alpha_0^{c_s}\right)^3 r}{c_s} \\
& + \frac{0.125 \left(\alpha_0^s\right)^3 r}{c_s} + \frac{0.0234375 \alpha_0^{c_s} r^3}{c_s^3} \\
& + \frac{0.0234375 \alpha_0^s r^3}{c_s^3} + \frac{0.109375 \left(\alpha_0^{c_s}\right)^2 r^2}{c_s^2} \\
& + \frac{0.109375 \left(\alpha_0^s\right)^2 r^2}{c_s^2} - \frac{0.0625 \alpha_1^{c_s} r^2}{c_s^2} \\
& - \frac{0.0625 \alpha_1^s r^2}{c_s^2} + \frac{0.375 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r}{c_s} \\
& + \frac{0.375 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r}{c_s} + \frac{0.21875 \alpha_0^{c_s} \alpha_0^s r^2}{c_s^2} \\
& - \frac{0.375 \alpha_0^{c_s} \alpha_1^{c_s} r}{c_s} - \frac{0.375 \alpha_0^{c_s} \alpha_1^s r}{c_s} \\
& - \frac{0.375 \alpha_0^s \alpha_1^{c_s} r}{c_s} - \frac{0.375 \alpha_0^s \alpha_1^s r}{c_s}
\end{aligned}$$

**Absolute order 5.**

$$\begin{aligned}
R_{3,5} = {} & - \frac{0.000762939453125 r^5}{c_s^5} + \frac{0.125 \alpha_3^s r}{c_s} \\
& + \frac{0.0419921875 \alpha_1^s r^3}{c_s^3} + \frac{0.28379535568268 \alpha_3^{c_s} r}{c_s} \\
& + \frac{0.071766316690503 \alpha_1^{c_s} r^3}{c_s^3} - \frac{0.125 \left(\alpha_0^s\right)^4 r}{c_s} \\
& - \frac{0.375 \left(\alpha_1^s\right)^2 r}{c_s} - \frac{0.078125 \alpha_2^s r^2}{c_s^2} \\
& - \frac{0.28125 \left(\alpha_0^s\right)^3 r^2}{c_s^2} - \frac{0.01708984375 \alpha_0^s r^4}{c_s^4} \\
& - \frac{0.1201171875 \left(\alpha_0^s\right)^2 r^3}{c_s^3} - \frac{0.28379535568268 \left(\alpha_0^{c_s}\right)^4 r}{c_s} \\
& - \frac{0.15752267784134 \alpha_2^{c_s} r^2}{c_s^2} - \frac{0.85138606704805 \left(\alpha_1^{c_s}\right)^2 r}{c_s} \\
& - \frac{0.024533376047626 \alpha_0^{c_s} r^4}{c_s^4} - \frac{0.5591418724447 \left(\alpha_0^{c_s}\right)^3 r^2}{c_s^2} \\
& - \frac{0.20943957507151 \left(\alpha_0^{c_s}\right)^2 r^3}{c_s^3} \\
& + \frac{0.0625 \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2} \\
& + \frac{0.125 \left(\alpha_0^s\right)^4 r \log\left(c_s\right)}{c_s}
\end{aligned}$$

$$
\begin{aligned}
&+ \frac{0.005859375\alpha_0^s r^4 \log\left(c_s\right)}{c_s^4} \\
&+ \frac{0.375\left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.75\left(\alpha_0^s\right)^2 \alpha_1^s r}{c_s} \\
&+ \frac{0.21875\left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.4375\alpha_0^s \alpha_1^s r^2}{c_s^2} \\
&+ \frac{0.0703125\left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{0.72129535568268\alpha_0^{c_s} \alpha_1^s r^2}{c_s^2} \\
&+ \frac{1.7027721340961\left(\alpha_0^{c_s}\right)^2 \alpha_1^{c_s} r}{c_s} \\
&+ \frac{1.6013860670481\left(\alpha_0^s\right)^2 \alpha_1^{c_s} r}{c_s} \\
&+ \frac{1.9763860670481\left(\alpha_0^{c_s}\right)^2 \alpha_1^s r}{c_s} \\
&+ \frac{0.7778918724447\alpha_0^s \alpha_1^{c_s} r^2}{c_s^2} \\
&+ \frac{0.87418722812738\alpha_0^{c_s} \alpha_1^{c_s} r^2}{c_s^2} - \frac{0.125\alpha_3^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.5\alpha_0^s \alpha_2^s r}{c_s} - \frac{0.0234375\alpha_1^s r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{1.1351814227307\alpha_0^{c_s} \alpha_2^{c_s} r}{c_s} - \frac{1.0337953556827\alpha_0^{c_s} \alpha_2^s r}{c_s} \\
&- \frac{1.0337953556827\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r}{c_s} \\
&- \frac{1.1013860670481\alpha_0^s \alpha_2^{c_s} r}{c_s} \\
&- \frac{1.3513860670481\left(\alpha_0^{c_s}\right)^3 \alpha_0^s r}{c_s} \\
&- \frac{1.6013860670481\alpha_1^{c_s} \alpha_1^s r}{c_s} \\
&- \frac{1.9763860670481\left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r}{c_s} \\
&- \frac{1.6182837448894\left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^2}{c_s^2} \\
&- \frac{1.3403918724447\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^2}{c_s^2} \\
&- \frac{0.35299426257151\alpha_0^{c_s} \alpha_0^s r^3}{c_s^3} \\
&+ \frac{0.125\alpha_0^s \alpha_2^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.125\left(\alpha_0^{c_s}\right)^3 \alpha_0^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{0.5\alpha_0^s \alpha_2^s r \log\left(c_s\right)}{c_s}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{0.375\alpha_0^{c_s}\alpha_2^s r\log(c_s)}{c_s} \\
&+ \frac{0.375\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r\log(c_s)}{c_s} \\
&+ \frac{0.375\alpha_1^{c_s}\alpha_1^s r\log(c_s)}{c_s} \\
&+ \frac{0.375\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r\log(c_s)}{c_s} \\
&+ \frac{0.21875\left(\alpha_0^{c_s}\right)^2\alpha_0^s r^2\log(c_s)}{c_s^2} \\
&+ \frac{0.4375\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^2\log(c_s)}{c_s^2} \\
&+ \frac{0.0703125\alpha_0^{c_s}\alpha_0^s r^3\log(c_s)}{c_s^3} \\
&+ \frac{3.1013860670481\alpha_0^{c_s}\alpha_0^s\alpha_1^s r}{c_s} \\
&+ \frac{3.6791582011442\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s} r}{c_s} \\
&- \frac{0.125\alpha_0^s\alpha_1^{c_s} r^2\log(c_s)}{c_s^2} \\
&- \frac{0.375\left(\alpha_0^s\right)^2\alpha_1^{c_s} r\log(c_s)}{c_s} \\
&- \frac{0.375\left(\alpha_0^{c_s}\right)^2\alpha_1^s r\log(c_s)}{c_s} \\
&- \frac{0.75\left(\alpha_0^s\right)^2\alpha_1^s r\log(c_s)}{c_s} \\
&- \frac{0.21875\alpha_0^{c_s}\alpha_1^s r^2\log(c_s)}{c_s^2} \\
&- \frac{0.34375\alpha_0^s\alpha_1^s r^2\log(c_s)}{c_s^2} \\
&- \frac{0.375\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s} r\log(c_s)}{c_s} \\
&- \frac{1.125\alpha_0^{c_s}\alpha_0^s\alpha_1^s r\log(c_s)}{c_s}
\end{aligned}
$$

**Absolute order 6.**

$$
\begin{aligned}
R_{3,6} = &\frac{0.00069887861331449 r^6}{c_s^6} + \frac{0.125\alpha_4^s r}{c_s} \\
&+ \frac{1.065570626349\left(\alpha_0^{c_s}\right)^3 r^3}{c_s^3} \\
&+ \frac{1.4176370369568\left(\alpha_0^{c_s}\right)^4 r^2}{c_s^2} \\
&+ \frac{0.52264594794504\left(\alpha_1^s\right)^2 r^2}{c_s^2} + \frac{0.10280837917801\left(\alpha_0^s\right)^5 r}{c_s} \\
&+ \frac{0.20825502603924\left(\alpha_0^{c_s}\right)^5 r}{c_s} \\
&+ \frac{0.24809448714071\left(\alpha_0^{c_s}\right)^2 r^4}{c_s^4} \\
&+ \frac{0.52433019631832\left(\alpha_0^s\right)^3 r^3}{c_s^3} + \frac{0.13631419746067\alpha_2^{c_s} r^3}{c_s^3}
\end{aligned}
$$

$$+\frac{0.017257388611671\alpha_0^s r^5}{c_s^5}+\frac{1.1501885493287\left(\alpha_1^{c_s}\right)^2 r^2}{c_s^2}$$
$$+\frac{0.073695194469 16\alpha_2^s r^3}{c_s^3}+\frac{0.1495700174229\left(\alpha_0^s\right)^2 r^4}{c_s^4}$$
$$+\frac{0.61990642191755\left(\alpha_0^s\right)^4 r^2}{c_s^2}+\frac{0.33325502603924\alpha_4^{c_s} r}{c_s}$$
$$+\frac{0.022103438284604\alpha_0^{c_s} r^5}{c_s^5}-\frac{0.042328393296221\alpha_1^s r^4}{c_s^4}$$
$$-\frac{0.066357117878927\alpha_1^{c_s} r^4}{c_s^4}-\frac{0.1038270947945\alpha_3^s r^2}{c_s^2}$$
$$-\frac{0.22780402727446\alpha_3^{c_s} r^2}{c_s^2}+\frac{0.0625\alpha_4^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.0625\left(\alpha_0^s\right)^5 r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.01171875\alpha_2^s r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{0.01171875\left(\alpha_0^s\right)^2 r^4\log\left(c_s\right)^2}{c_s^4}$$
$$+\frac{0.078125\alpha_3^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.21875\left(\alpha_0^s\right)^4 r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.171875\left(\alpha_1^s\right)^2 r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.10546875\left(\alpha_0^s\right)^3 r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{0.01708984375\alpha_1^s r^4\log\left(c_s\right)}{c_s^4}$$
$$+\frac{4.5746925879539\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^2 r}{c_s}$$
$$+\frac{1.1352294412117\alpha_0^{c_s}\alpha_2^s r^2}{c_s^2}$$
$$+\frac{2.6219204538578\left(\alpha_0^s\right)^2\alpha_2^{c_s} r}{c_s}$$
$$+\frac{3.8874132271834\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^3 r}{c_s}$$
$$+\frac{2.2571479479676\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^3}{c_s^3}$$
$$+\frac{1.9409334773531\left(\alpha_0^{c_s}\right)^4\alpha_0^s r}{c_s}$$
$$+\frac{3.5111972716302\alpha_0^{c_s}\left(\alpha_1^s\right)^2 r}{c_s}$$
$$+\frac{8.2127818560984\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r^2}{c_s^2}$$
$$+\frac{4.4950282979633\alpha_0^s\left(\alpha_1^{c_s}\right)^2 r}{c_s}$$
$$+\frac{1.6087005501362\alpha_0^s\left(\alpha_1^s\right)^2 r}{c_s}$$

$$
\begin{aligned}
&+ \frac{1.0946586542461 \left(\alpha_0^s\right)^2 \alpha_2^s r}{c_s} \\
&+ \frac{0.71467513753404 \alpha_0^s \alpha_2^s r^2}{c_s^2} \\
&+ \frac{1.6030226512749 \alpha_0^{c_s} \alpha_2^{c_s} r^2}{c_s^2} \\
&+ \frac{0.40191656771049 \alpha_0^{c_s} \alpha_0^s r^4}{c_s^4} \\
&+ \frac{1.1482074697214 \alpha_0^{c_s} \left(\alpha_0^s\right)^4 r}{c_s} \\
&+ \frac{2.4575502603924 \left(\alpha_0^{c_s}\right)^2 \alpha_2^{c_s} r}{c_s} \\
&+ \frac{1.4228234146217 \alpha_0^s \alpha_2^{c_s} r^2}{c_s^2} \\
&+ \frac{2.8877107655697 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^3}{c_s^3} \\
&+ \frac{6.1460976449703 \left(\alpha_0^{c_s}\right)^3 \alpha_0^s r^2}{c_s^2} \\
&+ \frac{3.4988253905886 \alpha_0^{c_s} \left(\alpha_1^{c_s}\right)^2 r}{c_s} \\
&+ \frac{1.8602805292847 \alpha_1^{c_s} \alpha_1^s r^2}{c_s^2} \\
&+ \frac{3.9539880896494 \left(\alpha_0^{c_s}\right)^2 \alpha_2^s r}{c_s} \\
&+ \frac{3.9948526700024 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 r^2}{c_s^2} \\
&- \frac{0.03125 \alpha_3^s r^2 \log\left(c_s\right)^2}{c_s^2} - \frac{0.125 \alpha_4^s r \log\left(c_s\right)}{c_s} \\
&- \frac{0.125 \left(\alpha_0^s\right)^5 r \log\left(c_s\right)}{c_s} \\
&- \frac{0.0029296875 \alpha_1^s r^4 \log\left(c_s\right)^2}{c_s^4} \\
&- \frac{0.4375 \left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.5625 \left(\alpha_0^s\right)^4 r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.068359375 \left(\alpha_0^s\right)^2 r^4 \log\left(c_s\right)}{c_s^4} \\
&- \frac{0.0419921875 \alpha_2^s r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{0.003814697265625 \alpha_0^s r^5 \log\left(c_s\right)}{c_s^5} \\
&- \frac{0.3603515625 \left(\alpha_0^s\right)^3 r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{0.57385180946901 \alpha_0^s \alpha_1^s r^3}{c_s^3} \\
&- \frac{2.4110271601353 \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r}{c_s}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{1.2147823321874\alpha_0^{c_s}\alpha_3^s r}{c_s} - \frac{1.4645474103051\alpha_0^s\alpha_3^{c_s} r}{c_s} \\
&- \frac{2.6441120746798\alpha_1^s\alpha_2^{c_s} r}{c_s} \\
&- \frac{4.5968842087759\left(\alpha_0^{c_s}\right)^3\alpha_1^s r}{c_s} \\
&- \frac{1.9544879813691\left(\alpha_0^s\right)^2\alpha_1^s r^2}{c_s^2} \\
&- \frac{4.4382967251885\left(\alpha_0^{c_s}\right)^2\alpha_1^{c_s} r^2}{c_s^2} \\
&- \frac{1.1168502750681\alpha_1^s\alpha_2^s r}{c_s} - \frac{1.1225357717973\alpha_0^{c_s}\alpha_1^{c_s} r^3}{c_s^3} \\
&- \frac{0.81325404477083\alpha_0^{c_s}\alpha_1^s r^3}{c_s^3} \\
&- \frac{1.0502754126021\left(\alpha_0^s\right)^3\alpha_1^s r}{c_s} \\
&- \frac{4.3255592153975\left(\alpha_0^{c_s}\right)^2\alpha_1^s r^2}{c_s^2} \\
&- \frac{2.2075502603924\left(\alpha_0^{c_s}\right)^3\alpha_1^{c_s} r}{c_s} \\
&- \frac{0.58061675835603\alpha_0^s\alpha_3^s r}{c_s} - \frac{2.4776020226013\alpha_1^{c_s}\alpha_2^s r}{c_s} \\
&- \frac{2.5825502603924\alpha_1^{c_s}\alpha_2^{c_s} r}{c_s} - \frac{1.4162751301962\alpha_0^{c_s}\alpha_3^{c_s} r}{c_s} \\
&- \frac{3.5073089371114\left(\alpha_0^s\right)^2\alpha_1^{c_s} r^2}{c_s^2} \\
&- \frac{0.9442318016501\alpha_0^s\alpha_1^{c_s} r^3}{c_s^3} \\
&+ \frac{0.0625\alpha_1^{c_s}\alpha_1^s r^2\log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.0625\left(\alpha_0^s\right)^2\alpha_2^{c_s} r\log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.0625\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^2 r\log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.1875\alpha_0^{c_s}\left(\alpha_0^s\right)^4 r\log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.1875\left(\alpha_0^{c_s}\right)^2\alpha_2^s r\log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.1875\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^3 r\log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.625\alpha_0^s\alpha_3^s r\log\left(c_s\right)}{c_s} \\
&+ \frac{0.625\left(\alpha_0^s\right)^2\alpha_2^s r\log\left(c_s\right)^2}{c_s} \\
&+ \frac{1.25\alpha_1^s\alpha_2^s r\log\left(c_s\right)}{c_s} \\
&+ \frac{1.25\left(\alpha_0^s\right)^3\alpha_1^s r\log\left(c_s\right)}{c_s} \\
&+ \frac{0.109375\alpha_0^{c_s}\alpha_2^s r^2\log\left(c_s\right)^2}{c_s^2}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{0.21875 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.4375 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.5625 \alpha_0^{c_s} \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.234375 \alpha_0^s \alpha_2^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.9375 \alpha_0^s \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.10546875 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)^2}{c_s^3} \\
&+ \frac{1.71875 \left(\alpha_0^s\right)^2 \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.3662109375 \alpha_0^s \alpha_1^s r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{0.28379535568268 \alpha_0^s \alpha_3^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{1.0337953556827 \alpha_0^{c_s} \alpha_3^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{1.1013860670481 \alpha_1^s \alpha_2^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{1.3513860670481 \left(\alpha_0^{c_s}\right)^3 \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{1.6013860670481 \alpha_1^{c_s} \alpha_2^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{1.6013860670481 \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{1.5557837448894 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{6.243371063951 \alpha_0^{c_s} \alpha_0^s \alpha_2^{c_s} r}{c_s} \\
&+ \frac{1.6182837448894 \left(\alpha_0^{c_s}\right)^2 \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{0.21529895007151 \alpha_0^s \alpha_1^{c_s} r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{4.7259796038175 \alpha_0^{c_s} \alpha_0^s \alpha_2^s r}{c_s} \\
&+ \frac{0.35299426257151 \alpha_0^{c_s} \alpha_1^s r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{7.299656342872 \alpha_0^s \alpha_1^{c_s} \alpha_1^s r}{c_s} \\
&+ \frac{10.486494425184 \alpha_0^{c_s} \alpha_1^{c_s} \alpha_1^s r}{c_s} \\
&- \frac{0.0625 \alpha_1^s \alpha_2^{c_s} r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.0625 \left(\alpha_0^{c_s}\right)^3 \alpha_1^s r \log\left(c_s\right)^2}{c_s}
\end{aligned}
$$

$$- \frac{0.125 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{0.1875 \alpha_0^{c_s} \alpha_3^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.1875 \alpha_1^{c_s} \alpha_2^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.1875 \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.3125 \alpha_0^s \alpha_3^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.625 \alpha_1^s \alpha_2^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.625 \left(\alpha_0^s\right)^3 \alpha_1^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{1.25 \left(\alpha_0^s\right)^2 \alpha_2^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.109375 \left(\alpha_0^{c_s}\right)^2 \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{0.03515625 \alpha_0^{c_s} \alpha_1^s r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{1.875 \alpha_0^s \left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.59375 \alpha_0^s \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{0.10546875 \alpha_0^s \alpha_1^s r^3 \log\left(c_s\right)^2}{c_s^3}$$
$$- \frac{0.671875 \left(\alpha_0^s\right)^2 \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{0.28379535568268 \left(\alpha_0^{c_s}\right)^4 \alpha_0^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.31504535568268 \alpha_0^s \alpha_2^{c_s} r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{0.72129535568268 \alpha_0^{c_s} \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{0.85138606704805 \alpha_0^s \left(\alpha_1^{c_s}\right)^2 r \log\left(c_s\right)}{c_s}$$
$$- \frac{1.0337953556827 \alpha_0^{c_s} \left(\alpha_0^s\right)^4 r \log\left(c_s\right)}{c_s}$$
$$- \frac{1.1013860670481 \left(\alpha_0^s\right)^2 \alpha_2^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{1.3513860670481 \left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^2 r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.098133504190503 \alpha_0^{c_s} \alpha_0^s r^4 \log\left(c_s\right)}{c_s^4}$$
$$- \frac{1.9763860670481 \left(\alpha_0^{c_s}\right)^2 \alpha_2^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{1.9763860670481 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^3 r \log\left(c_s\right)}{c_s}$$

$$
\begin{aligned}
&- \frac{1.1182837448894 \left(\alpha_0^{c_s}\right)^3 \alpha_0^s r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{3.1013860670481 \alpha_0^{c_s} \left(\alpha_1^s\right)^2 r \log\left(c_s\right)}{c_s} \\
&- \frac{0.7778918724447 \alpha_1^{c_s} \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{3.2365674897788 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{2.6807837448894 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.62831872521453 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{10.317828730023 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^{c_s} r}{c_s} \\
&- \frac{10.419919562718 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r}{c_s} \\
&- \frac{6.2653115524259 \alpha_0^{c_s} \alpha_0^s \alpha_1^s r^2}{c_s^2} \\
&- \frac{1.0589827877145 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{11.728814544016 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^s r}{c_s} \\
&- \frac{8.7476479574518 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} r^2}{c_s^2} \\
&- \frac{6.9558196807943 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^s r}{c_s} \\
&+ \frac{0.1875 \alpha_0^{c_s} \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.75 \alpha_0^{c_s} \alpha_0^s \alpha_2^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.5625 \alpha_0^s \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{1.7027721340961 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{6.2027721340961 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{1.7483744562548 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} r^2 \log\left(c_s\right)}{c_s^2} \\
&+ \frac{3.6791582011442 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r \log\left(c_s\right)}{c_s} \\
&+ \frac{5.9291582011442 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&+ \frac{4.1233744562548 \alpha_0^{c_s} \alpha_0^s \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{0.1875 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.5625 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^s r \log\left(c_s\right)^2}{c_s}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{1.125\alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^s r \log\left(c_s\right)^2}{c_s} \\
&- \frac{0.65625\alpha_0^{c_s} \alpha_0^s \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2} \\
&- \frac{1.1351814227307\alpha_0^{c_s} \alpha_0^s \alpha_2^{c_s} r \log\left(c_s\right)}{c_s} \\
&- \frac{4.1351814227307\alpha_0^{c_s} \alpha_0^s \alpha_2^s r \log\left(c_s\right)}{c_s} \\
&- \frac{3.6791582011442\alpha_0^{c_s} \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&- \frac{4.8041582011442\alpha_0^s \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)}{c_s}
\end{aligned}
$$

**Absolute order 7.**

$$
\begin{aligned}
R_{3,7} = &- \frac{0.00066040799045703 r^7}{c_s^7} + \frac{0.125\alpha_5^s r}{c_s} \\
&+ \frac{1.2557055991705 \left(\alpha_1^{c_s}\right)^3 r}{c_s} + \frac{0.36074999115751\alpha_5^{c_s} r}{c_s} \\
&+ \frac{0.044182253542082\alpha_1^s r^5}{c_s^5} + \frac{0.10547078563903\alpha_3^s r^3}{c_s^3} \\
&+ \frac{0.067614788198182\alpha_1^{c_s} r^5}{c_s^5} + \frac{1.0840862194786 \left(\alpha_1^s\right)^3 r}{c_s} \\
&+ \frac{0.22086108655179\alpha_3^{c_s} r^3}{c_s^3} - \frac{0.076954635474829 \left(\alpha_0^{c_s}\right)^6 r}{c_s} \\
&- \frac{0.12260923390637\alpha_4^s r^2}{c_s^2} - \frac{0.065244100954277 \left(\alpha_0^s\right)^6 r}{c_s} \\
&- \frac{0.31475559837547 \left(\alpha_0^{c_s}\right)^2 r^5}{c_s^5} \\
&- \frac{1.8011431833565 \left(\alpha_0^s\right)^4 r^3}{c_s^3} \\
&- \frac{1.8035217754453 \left(\alpha_0^{c_s}\right)^3 r^4}{c_s^4} \\
&- \frac{1.1197437723225 \left(\alpha_0^s\right)^5 r^2}{c_s^2} - \frac{0.2991670940175\alpha_4^{c_s} r^2}{c_s^2} \\
&- \frac{0.076712092161027\alpha_2^s r^4}{c_s^4} - \frac{0.70760825653778 \left(\alpha_1^s\right)^2 r^3}{c_s^3} \\
&- \frac{0.19445546235151 \left(\alpha_0^s\right)^2 r^5}{c_s^5} \\
&- \frac{4.1620983497444 \left(\alpha_0^{c_s}\right)^4 r^3}{c_s^3} - \frac{0.023975376743708\alpha_0^{c_s} r^6}{c_s^6} \\
&- \frac{0.13853540110911\alpha_2^{c_s} r^4}{c_s^4} - \frac{1.154727777479 \left(\alpha_2^{c_s}\right)^2 r}{c_s} \\
&- \frac{0.018648400200653\alpha_0^s r^6}{c_s^6} - \frac{2.675269515726 \left(\alpha_0^{c_s}\right)^5 r^2}{c_s^2} \\
&- \frac{1.5385686369455 \left(\alpha_1^{c_s}\right)^2 r^3}{c_s^3} \\
&- \frac{0.81762161375293 \left(\alpha_2^s\right)^2 r}{c_s} - \frac{0.9077746502428 \left(\alpha_0^s\right)^3 r^4}{c_s^4} \\
&+ \frac{0.0009765625\alpha_2^s r^4 \log\left(c_s\right)^3}{c_s^4}
\end{aligned}
$$

$$
\begin{aligned}
&+ \frac{0.015625 \left(\alpha_0^s\right)^3 r^4 \log\left(c_s\right)^3}{c_s^4} \\
&+ \frac{0.0625 \alpha_5^s r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.03515625 \left(\alpha_1^s\right)^2 r^3 \log\left(c_s\right)^3}{c_s^3} \\
&+ \frac{0.9375 \left(\alpha_1^s\right)^3 r \log\left(c_s\right)^2}{c_s} \\
&+ \frac{0.10546875 \left(\alpha_0^s\right)^4 r^3 \log\left(c_s\right)^3}{c_s^3} \\
&+ \frac{0.02099609375 \alpha_3^s r^3 \log\left(c_s\right)^2}{c_s^3} \\
&+ \frac{0.0019073486328125 \alpha_1^s r^5 \log\left(c_s\right)^2}{c_s^5} \\
&+ \frac{3.9387181395716 \alpha_1^{c_s} \alpha_2^{c_s} r^2}{c_s^2} \\
&+ \frac{0.57385180946901 \left(\alpha_1^s\right)^2 r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{10.09950678112 \left(\alpha_1^{c_s}\right)^2 \alpha_1^s r}{c_s} \\
&+ \frac{1.4409653155906 \alpha_0^{c_s} \alpha_1^{c_s} r^4}{c_s^4} \\
&+ \frac{2.2829930145581 \alpha_0^{c_s} \alpha_3^{c_s} r^2}{c_s^2} \\
&+ \frac{6.0506877923487 \left(\alpha_0^{c_s}\right)^4 \alpha_1^s r}{c_s} \\
&+ \frac{0.75724388321833 \alpha_0^s \alpha_1^s r^4}{c_s^4} \\
&+ \frac{1.1881148878051 \left(\alpha_0^{c_s}\right)^4 \alpha_1^{c_s} r}{c_s} \\
&+ \frac{1.2633539166158 \left(\alpha_0^s\right)^2 \alpha_3^s r}{c_s} \\
&+ \frac{10.84818865722 \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r^2}{c_s^2} \\
&+ \frac{0.086286943058355 \left(\alpha_0^s\right)^2 r^5 \log\left(c_s\right)}{c_s^5} \\
&+ \frac{5.8694148411763 \left(\alpha_0^s\right)^3 \alpha_1^s r^2}{c_s^2} \\
&+ \frac{3.899202830862 \left(\alpha_0^s\right)^2 \alpha_1^s r^3}{c_s^3} \\
&+ \frac{1.0138030827023 \left(\alpha_0^s\right)^4 \alpha_1^s r}{c_s} \\
&+ \frac{1.0488038466557 \alpha_0^{c_s} \alpha_1^s r^4}{c_s^4} \\
&+ \frac{2.1070916662185 \left(\alpha_0^{c_s}\right)^2 \alpha_3^{c_s} r}{c_s} \\
&+ \frac{0.004193271679887 \alpha_0^s r^6 \log\left(c_s\right)}{c_s^6}
\end{aligned}
$$

$$+\frac{8.7674965667948\left(\alpha_0^{c_s}\right)^2\alpha_1^{c_s}r^3}{c_s^3}$$
$$+\frac{1.1168502750681\left(\alpha_2^s\right)^2 r\log\left(c_s\right)}{c_s}$$
$$+\frac{5.0351500609997\left(\alpha_0^{c_s}\right)^2\alpha_3^s r}{c_s}$$
$$+\frac{0.042328393296221\alpha_2^s r^4\log\left(c_s\right)}{c_s^4}$$
$$+\frac{6.5251062314093\left(\alpha_0^s\right)^2\alpha_1^{c_s}r^3}{c_s^3}$$
$$+\frac{0.10280837917801\left(\alpha_0^s\right)^6 r\log\left(c_s\right)}{c_s}$$
$$+\frac{7.5978803659804\left(\alpha_0^{c_s}\right)^2\alpha_1^s r^3}{c_s^3}$$
$$+\frac{17.170223355941\left(\alpha_0^{c_s}\right)^3\alpha_1^s r^2}{c_s^2}$$
$$+\frac{2.1520321750819\alpha_0^s\alpha_3^{c_s}r^2}{c_s^2}$$
$$+\frac{0.14583333333333\left(\alpha_0^s\right)^5 r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{0.5982800696916\left(\alpha_0^s\right)^3 r^4\log\left(c_s\right)}{c_s^4}$$
$$+\frac{1.200524716176\alpha_0^s\alpha_1^{c_s}r^4}{c_s^4}$$
$$+\frac{2.4730563846109\left(\alpha_0^s\right)^4\alpha_1^{c_s}r}{c_s}$$
$$+\frac{1.2398128438351\left(\alpha_0^s\right)^5 r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.010416666666667\alpha_4^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{0.020833333333333\left(\alpha_0^s\right)^6 r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{3.1420899004122\alpha_1^{c_s}\alpha_2^s r^2}{c_s^2}$$
$$+\frac{1.572990588955\left(\alpha_0^s\right)^4 r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{1.5760620651767\alpha_0^{c_s}\alpha_3^s r^2}{c_s^2}$$
$$+\frac{3.2941625300387\left(\alpha_0^s\right)^2\alpha_3^{c_s}r}{c_s}$$
$$+\frac{0.1038270947945\alpha_4^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.20833333333333\left(\alpha_2^s\right)^2 r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{1.7276736271957\alpha_1^s\alpha_2^s r^2}{c_s^2}$$
$$+\frac{13.88289571595\left(\alpha_0^{c_s}\right)^3\alpha_1^{c_s}r^2}{c_s^2}$$

$$+\frac{3.51534396509\alpha_1^s\alpha_2^{c_s}r^2}{c_s^2}+\frac{0.96375532208646\alpha_0^s\alpha_3^sr^2}{c_s^2}$$
$$+\frac{7.7423178156474\alpha_1^{c_s}\left(\alpha_1^s\right)^2r}{c_s}$$
$$-\frac{0.00390625\alpha_3^sr^3\log\left(c_s\right)^3}{c_s^3}$$
$$-\frac{0.0625\left(\alpha_0^s\right)^6r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.125\alpha_5^sr\log\left(c_s\right)}{c_s}$$
$$-\frac{0.0390625\alpha_4^sr^2\log\left(c_s\right)^2}{c_s^2}$$
$$-\frac{0.3125\left(\alpha_1^s\right)^3r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.625\left(\alpha_2^s\right)^2r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.5625\left(\alpha_0^s\right)^5r^2\log\left(c_s\right)^2}{c_s^2}$$
$$-\frac{0.008544921875\alpha_2^sr^4\log\left(c_s\right)^2}{c_s^4}$$
$$-\frac{0.13671875\left(\alpha_0^s\right)^3r^4\log\left(c_s\right)^2}{c_s^4}$$
$$-\frac{0.18310546875\left(\alpha_1^s\right)^2r^3\log\left(c_s\right)^2}{c_s^3}$$
$$-\frac{0.0095367431640625\left(\alpha_0^s\right)^2r^5\log\left(c_s\right)^2}{c_s^5}$$
$$-\frac{0.54052734375\left(\alpha_0^s\right)^4r^3\log\left(c_s\right)^2}{c_s^3}$$

$$-\frac{1.4855117920869\alpha_0^{c_s}\alpha_2^sr^3}{c_s^3}-\frac{1.0111179362879\alpha_0^s\alpha_2^sr^3}{c_s^3}$$
$$-\frac{0.60587387125098\alpha_0^s\alpha_4^sr}{c_s}-\frac{2.3683440805453\alpha_1^{c_s}\alpha_1^sr^3}{c_s^3}$$
$$-\frac{3.1468339532713\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2r}{c_s}$$
$$-\frac{3.3445455614335\alpha_1^s\alpha_3^{c_s}r}{c_s}$$
$$-\frac{7.7328550642049\left(\alpha_0^{c_s}\right)^2\alpha_2^sr^2}{c_s^2}$$
$$-\frac{0.51503239853261\alpha_0^{c_s}\alpha_0^sr^5}{c_s^5}$$
$$-\frac{10.786415997547\alpha_0^{c_s}\left(\alpha_1^{c_s}\right)^2r^2}{c_s^2}$$
$$-\frac{11.101166727647\alpha_0^s\left(\alpha_1^{c_s}\right)^2r^2}{c_s^2}$$
$$-\frac{1.6087005501362\left(\alpha_1^s\right)^3r\log\left(c_s\right)}{c_s}$$
$$-\frac{27.261332258597\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^3r^2}{c_s^2}$$

$$
- \frac{7.6667459952952 \left(\alpha_0^{c_s}\right)^2 \alpha_2^{c_s} r^2}{c_s^2}
$$

$$
- \frac{3.897507710927 \alpha_2^{c_s} \alpha_2^s r}{c_s} - \frac{2.1408870219012 \alpha_1^{c_s} \alpha_3^{c_s} r}{c_s}
$$

$$
- \frac{2.1930683622079 \alpha_0^{c_s} \alpha_2^{c_s} r^3}{c_s^3}
$$

$$
- \frac{4.6296729998974 \alpha_0^s \left(\alpha_1^s\right)^2 r^2}{c_s^2}
$$

$$
- \frac{9.4629411206417 \alpha_0^{c_s} \left(\alpha_0^s\right)^4 r^2}{c_s^2}
$$

$$
- \frac{4.7412780427983 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s r^4}{c_s^4}
$$

$$
- \frac{10.6792492855 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 r^3}{c_s^3}
$$

$$
- \frac{1.3469092355797 \alpha_0^{c_s} \alpha_4^{c_s} r}{c_s}
$$

$$
- \frac{3.2598096829233 \left(\alpha_0^s\right)^2 \alpha_2^s r^2}{c_s^2}
$$

$$
- \frac{1.6656048141581 \alpha_0^s \alpha_4^{c_s} r}{c_s}
$$

$$
- \frac{3.7668956958108 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r^4}{c_s^4}
$$

$$
- \frac{7.8136689732526 \alpha_0^{c_s} \left(\alpha_1^s\right)^2 r^2}{c_s^2}
$$

$$
- \frac{16.377283754452 \left(\alpha_0^{c_s}\right)^3 \alpha_0^s r^3}{c_s^3}
$$

$$
- \frac{16.453977631147 \left(\alpha_0^{c_s}\right)^4 \alpha_0^s r^2}{c_s^2}
$$

$$
- \frac{32.974734898143 \left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^2 r^2}{c_s^2}
$$

$$
- \frac{8.4133427159559 \left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^3 r}{c_s}
$$

$$
- \frac{8.690822571652 \left(\alpha_0^{c_s}\right)^3 \alpha_2^s r}{c_s}
$$

$$
- \frac{0.017257388611671 \alpha_1^s r^5 \log\left(c_s\right)}{c_s^5}
$$

$$
- \frac{4.5715835447462 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^4 r}{c_s}
$$

$$
- \frac{0.07369519446916 \alpha_3^s r^3 \log\left(c_s\right)}{c_s^3}
$$

$$
- \frac{9.9483576869351 \left(\alpha_0^s\right)^2 \left(\alpha_1^{c_s}\right)^2 r}{c_s}
$$

$$
- \frac{1.2622310659043 \alpha_0^{c_s} \alpha_4^s r}{c_s} - \frac{1.2984954850039 \alpha_1^s \alpha_3^s r}{c_s}
$$

$$
- \frac{21.116091248333 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 r^3}{c_s^3}
$$

$$
- \frac{0.020833333333333 \alpha_5^s r \log\left(c_s\right)^3}{c_s}
$$

$$
\begin{aligned}
&- \frac{2.8700480383984\alpha_1^{c_s}\alpha_3^s r}{c_s} \\
&- \frac{1.459738851267\left(\alpha_0^{c_s}\right)^5\alpha_0^s r}{c_s} \\
&- \frac{1.4700626232957\left(\alpha_0^s\right)^3\alpha_2^s r}{c_s} \\
&- \frac{6.0003047609539\left(\alpha_0^{c_s}\right)^4\left(\alpha_0^s\right)^2 r}{c_s} \\
&- \frac{6.6279464500316\left(\alpha_0^s\right)^2\alpha_2^{c_s} r^2}{c_s^2} \\
&- \frac{14.104474158519\left(\alpha_0^{c_s}\right)^2\left(\alpha_1^s\right)^2 r}{c_s} \\
&- \frac{3.6644110968749\left(\alpha_0^s\right)^3\alpha_2^{c_s} r}{c_s} \\
&- \frac{3.6657307304634\left(\alpha_0^{c_s}\right)^2\left(\alpha_1^{c_s}\right)^2 r}{c_s} \\
&- \frac{1.9050809010702\alpha_0^s\alpha_2^{c_s} r^3}{c_s^3} \\
&- \frac{1.9242741322273\left(\alpha_0^{c_s}\right)^3\alpha_2^{c_s} r}{c_s} \\
&- \frac{0.96600547490636\alpha_0^{c_s}\left(\alpha_0^s\right)^5 r}{c_s} \\
&+ \frac{0.0625\alpha_0^{c_s}\alpha_4^s r\log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.0625\alpha_0^{c_s}\left(\alpha_0^s\right)^5 r\log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.0625\alpha_1^{c_s}\alpha_3^s r\log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.0625\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^4 r\log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.125\alpha_0^s\alpha_4^s r\log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.01171875\alpha_0^{c_s}\alpha_2^s r^3\log\left(c_s\right)^3}{c_s^3} \\
&+ \frac{0.046875\alpha_0^s\alpha_2^s r^3\log\left(c_s\right)^3}{c_s^3} \\
&+ \frac{0.1875\left(\alpha_0^{c_s}\right)^2\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.375\alpha_0^s\alpha_3^s r^2\log\left(c_s\right)^2}{c_s^2} \\
&+ \frac{0.3125\alpha_1^s\alpha_3^s r\log\left(c_s\right)^3}{c_s} \\
&+ \frac{0.21875\alpha_0^{c_s}\left(\alpha_1^s\right)^2 r^2\log\left(c_s\right)^3}{c_s^2} \\
&+ \frac{0.5625\alpha_0^s\left(\alpha_1^s\right)^2 r^2\log\left(c_s\right)^3}{c_s^2} \\
&+ \frac{0.9375\left(\alpha_0^s\right)^4\alpha_1^s r\log\left(c_s\right)^2}{c_s}
\end{aligned}
$$

$$+\frac{0.9375\left(\alpha_0^s\right)^2\alpha_3^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.9375\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.10546875\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r^3\log\left(c_s\right)^3}{c_s^3}$$
$$+\frac{0.734375\alpha_1^s\alpha_2^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{2.84375\left(\alpha_0^s\right)^3\alpha_1^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.1025390625\alpha_0^s\alpha_1^s r^4\log\left(c_s\right)^2}{c_s^4}$$
$$+\frac{1.08984375\left(\alpha_0^s\right)^2\alpha_1^s r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{4.5746925879539\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^3 r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.14189767784134\left(\alpha_0^{c_s}\right)^4\alpha_1^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.14189767784134\left(\alpha_0^s\right)^2\alpha_3^{c_s} r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.15752267784134\alpha_1^s\alpha_2^{c_s} r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{10.623428137137\alpha_0^{c_s}\alpha_1^s\alpha_2^s r}{c_s}$$
$$+\frac{0.36064767784134\alpha_0^{c_s}\alpha_3^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{0.42569303352403\left(\alpha_1^{c_s}\right)^2\alpha_1^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{4.5507541434397\alpha_0^s\alpha_1^s\alpha_2^s r}{c_s}$$
$$+\frac{2.6219204538578\left(\alpha_0^s\right)^3\alpha_2^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.049066752095252\alpha_0^{c_s}\alpha_1^s r^4\log\left(c_s\right)^2}{c_s^4}$$
$$+\frac{0.80069303352403\left(\alpha_0^s\right)^4\alpha_1^{c_s} r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{3.8874132271834\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^4 r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.98819303352403\left(\alpha_0^{c_s}\right)^2\alpha_3^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.5591418724447\left(\alpha_0^{c_s}\right)^3\alpha_1^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{1.2147823321874\alpha_0^{c_s}\alpha_4^s r\log\left(c_s\right)}{c_s}$$

$$+\frac{1.4645474103051\alpha_1^s\alpha_3^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.38894593622235\alpha_1^{c_s}\alpha_2^s r^2\log\left(c_s\right)^2}{c_s^2}$$

$$+\frac{1.5557837448894\left(\alpha_0^s\right)^3\alpha_1^{c_s}r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{25.492893672906\left(\alpha_0^{c_s}\right)^3\alpha_0^s\alpha_1^s r}{c_s}$$
$$+\frac{1.9409334773531\left(\alpha_0^{c_s}\right)^4\left(\alpha_0^s\right)^2 r\log\left(c_s\right)}{c_s}$$
$$+\frac{16.183034565821\alpha_0^{c_s}\alpha_0^s\alpha_1^{c_s}r^3}{c_s^3}$$
$$+\frac{2.4020791005721\alpha_1^{c_s}\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{2.6441120746798\alpha_2^{c_s}\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{5.6448611555529\alpha_0^{c_s}\alpha_0^s\alpha_3^s r}{c_s}$$
$$+\frac{6.7714438439027\alpha_0^{c_s}\left(\alpha_0^s\right)^3 r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{14.908599752858\alpha_0^{c_s}\alpha_1^s\alpha_2^{c_s}r}{c_s}$$
$$+\frac{16.425563712197\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^3 r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{4.4950282979633\left(\alpha_0^s\right)^2\left(\alpha_1^{c_s}\right)^2 r\log\left(c_s\right)}{c_s}$$
$$+\frac{4.5968842087759\left(\alpha_0^{c_s}\right)^3\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{9.8824373480888\alpha_0^{c_s}\left(\alpha_0^s\right)^3\alpha_1^s r}{c_s}$$
$$+\frac{0.31415936260727\left(\alpha_0^{c_s}\right)^2\alpha_1^s r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{0.32294842510727\left(\alpha_0^s\right)^2\alpha_1^{c_s}r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{11.201715574019\alpha_0^s\alpha_1^s\alpha_2^{c_s}r}{c_s}$$
$$+\frac{2.8352740739136\left(\alpha_0^{c_s}\right)^4\alpha_0^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{5.9080038194127\alpha_0^{c_s}\alpha_1^{c_s}\alpha_2^{c_s}r}{c_s}$$
$$+\frac{6.2653115524259\alpha_0^{c_s}\left(\alpha_1^s\right)^2 r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{8.6631322967092\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{4.9542678586284\alpha_0^s\left(\alpha_1^s\right)^2 r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{10.612365854046\alpha_0^s\alpha_1^{c_s}\alpha_2^s r}{c_s}$$
$$+\frac{42.618229821464\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_1^s r^2}{c_s^2}$$
$$+\frac{10.755281999685\alpha_0^s\alpha_1^{c_s}\alpha_2^{c_s}r}{c_s}$$
$$+\frac{11.559920669238\alpha_0^{c_s}\alpha_0^s\alpha_1^s r^3}{c_s^3}$$

$$+\frac{11.728814544016\left(\alpha_0^{c_s}\right)^2\left(\alpha_1^s\right)^2 r\log\left(c_s\right)}{c_s}$$
$$+\frac{3.196711879047\left(\alpha_0^{c_s}\right)^3\alpha_0^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{1.6076662708419\alpha_0^{c_s}\left(\alpha_0^s\right)^2 r^4\log\left(c_s\right)}{c_s^4}$$
$$+\frac{0.81325404477083\alpha_0^{c_s}\alpha_2^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{0.20825502603924\left(\alpha_0^{c_s}\right)^5\alpha_0^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{6.9165270668905\alpha_0^{c_s}\alpha_0^s\alpha_3^{c_s} r}{c_s}$$
$$+\frac{27.819224792758\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2\alpha_1^s r}{c_s}$$
$$+\frac{0.11051719142302\alpha_0^{c_s}\alpha_0^s r^5\log\left(c_s\right)}{c_s^5}$$
$$+\frac{29.668801756997\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^s r^2}{c_s^2}$$
$$+\frac{0.99237794856283\left(\alpha_0^{c_s}\right)^2\alpha_0^s r^4\log\left(c_s\right)}{c_s^4}$$
$$+\frac{16.166403940833\alpha_0^{c_s}\left(\alpha_0^s\right)^3\alpha_1^{c_s} r}{c_s}$$
$$+\frac{16.932268645455\alpha_0^{c_s}\alpha_1^{c_s}\alpha_2^s r}{c_s}$$
$$+\frac{2.1449340668482\left(\alpha_0^s\right)^3\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{4.3255592153975\left(\alpha_0^{c_s}\right)^2\alpha_2^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{1.1482074697214\alpha_0^{c_s}\left(\alpha_0^s\right)^5 r\log\left(c_s\right)}{c_s}$$
$$+\frac{2.3003770986574\alpha_0^s\left(\alpha_1^{c_s}\right)^2 r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.036458333333333\left(\alpha_0^{c_s}\right)^2\alpha_2^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{0.14583333333333\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^3 r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{0.29166666666667\alpha_0^{c_s}\left(\alpha_0^s\right)^4 r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{4.7595267879426\left(\alpha_0^s\right)^2\left(\alpha_1^s\right)^2 r\log\left(c_s\right)}{c_s}$$
$$+\frac{2.4776020226013\alpha_1^{c_s}\alpha_3^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.33325502603924\alpha_0^s\alpha_4^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{0.020833333333333\alpha_2^{c_s}\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.020833333333333\left(\alpha_0^s\right)^3\alpha_2^{c_s} r\log\left(c_s\right)^3}{c_s}$$

$$+\frac{0.020833333333333\left(\alpha_0^{c_s}\right)^3\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.020833333333333\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^3 r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{43.051818854924\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^{c_s}r^2}{c_s^2}$$
$$+\frac{0.70561675835603\alpha_0^s\alpha_4^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{2.8456468292435\left(\alpha_0^s\right)^2\alpha_2^{c_s}r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{47.050382883379\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_1^{c_s}r^2}{c_s^2}$$
$$+\frac{0.38020833333333\left(\alpha_0^s\right)^2\alpha_2^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{12.292195289941\left(\alpha_0^{c_s}\right)^3\left(\alpha_0^s\right)^2 r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.79493739287649\alpha_0^s\alpha_2^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{0.408942592382\alpha_0^s\alpha_2^{c_s}r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{13.178724319326\left(\alpha_0^{c_s}\right)^3\alpha_0^s\alpha_1^{c_s}r}{c_s}$$
$$+\frac{0.41666666666667\left(\alpha_0^s\right)^3\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{3.3838382564372\left(\alpha_0^s\right)^2\alpha_2^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{1.6974670334241\alpha_1^s\alpha_3^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{27.472885902749\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2\alpha_1^{c_s}r}{c_s}$$
$$+\frac{0.9442318016501\alpha_1^{c_s}\alpha_1^s r^3\log\left(c_s\right)}{c_s^3}$$
$$+\frac{7.9897053400048\alpha_0^{c_s}\left(\alpha_0^s\right)^4 r^2\log\left(c_s\right)}{c_s^2}$$
$$-\frac{0.0625\left(\alpha_0^s\right)^4\alpha_1^{c_s}r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.0625\left(\alpha_0^{c_s}\right)^2\alpha_3^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.01171875\alpha_0^s\alpha_1^s r^4\log\left(c_s\right)^3}{c_s^4}$$
$$-\frac{0.1875\alpha_1^{c_s}\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.375\alpha_0^s\alpha_4^s r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{0.3125\left(\alpha_0^s\right)^4\alpha_1^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.3125\left(\alpha_0^s\right)^2\alpha_3^s r\log\left(c_s\right)^3}{c_s}$$

$$- \frac{1.25 \left(\alpha_0^s\right)^3 \alpha_2^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.9375 \alpha_1^s \alpha_3^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.2109375 \left(\alpha_0^s\right)^2 \alpha_1^s r^3 \log\left(c_s\right)^3}{c_s^3}$$
$$- \frac{2.8125 \left(\alpha_0^s\right)^2 \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.24609375 \alpha_0^s \alpha_2^s r^3 \log\left(c_s\right)^2}{c_s^3}$$

$$- \frac{2.15625 \alpha_0^s \left(\alpha_1^s\right)^2 r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{1.453125 \left(\alpha_0^s\right)^2 \alpha_2^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{0.14189767784134 \alpha_1^s \alpha_3^{c_s} r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.14189767784134 \left(\alpha_0^{c_s}\right)^4 \left(\alpha_0^s\right)^2 r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.31504535568268 \left(\alpha_0^s\right)^2 \alpha_2^{c_s} r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{0.42569303352403 \left(\alpha_0^s\right)^2 \left(\alpha_1^{c_s}\right)^2 r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.51689767784134 \alpha_0^{c_s} \alpha_4^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.51689767784134 \alpha_0^{c_s} \left(\alpha_0^s\right)^5 r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.55069303352403 \alpha_2^{c_s} \alpha_2^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.55069303352403 \left(\alpha_0^s\right)^3 \alpha_2^{c_s} r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{1.1352294412117 \alpha_0^{c_s} \alpha_3^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{0.67569303352403 \left(\alpha_0^{c_s}\right)^3 \alpha_2^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.67569303352403 \left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^3 r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.19626700838101 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 r^4 \log\left(c_s\right)^2}{c_s^4}$$
$$- \frac{0.80069303352403 \alpha_1^{c_s} \alpha_3^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{0.98819303352403 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^4 r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{1.1182837448894 \left(\alpha_0^{c_s}\right)^3 \left(\alpha_0^s\right)^2 r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{2.4110271601353 \left(\alpha_0^s\right)^4 \alpha_1^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{1.4645474103051 \left(\alpha_0^s\right)^2 \alpha_3^{c_s} r \log\left(c_s\right)}{c_s}$$

$$
- \frac{0.8091418724447\left(\alpha_0^{c_s}\right)^2\alpha_2^s r^2\log\left(c_s\right)^2}{c_s^2}
- \frac{3.2365674897788\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^3 r^2\log\left(c_s\right)^2}{c_s^2}
- \frac{0.10764947503576\alpha_1^{c_s}\alpha_1^s r^3\log\left(c_s\right)^2}{c_s^3}
- \frac{1.9409334773531\left(\alpha_0^{c_s}\right)^4\alpha_1^s r\log\left(c_s\right)}{c_s}
- \frac{2.6807837448894\alpha_0^{c_s}\left(\alpha_0^s\right)^4 r^2\log\left(c_s\right)^2}{c_s^2}
- \frac{10.851732081256\alpha_0^{c_s}\alpha_0^s\alpha_2^s r^2}{c_s^2}
- \frac{0.17649713128576\alpha_0^{c_s}\alpha_2^s r^3\log\left(c_s\right)^2}{c_s^3}
- \frac{2.9645791005721\left(\alpha_0^{c_s}\right)^2\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^2}{c_s}
- \frac{3.294546017362\left(\alpha_0^s\right)^2\alpha_1^s r^3\log\left(c_s\right)}{c_s^3}
- \frac{27.775184091118\left(\alpha_0^{c_s}\right)^2\alpha_1^{c_s}\alpha_1^s r}{c_s}
- \frac{2.0616872281274\alpha_0^{c_s}\left(\alpha_1^s\right)^2 r^2\log\left(c_s\right)^2}{c_s^2}
- \frac{4.4950282979633\left(\alpha_1^{c_s}\right)^2\alpha_1^s r\log\left(c_s\right)}{c_s}
- \frac{22.554554719686\alpha_0^{c_s}\alpha_1^{c_s}\alpha_1^s r^2}{c_s^2}
- \frac{7.299656342872\alpha_1^{c_s}\left(\alpha_1^s\right)^2 r\log\left(c_s\right)}{c_s}
- \frac{15.812179043725\alpha_0^{c_s}\alpha_0^s\alpha_2^{c_s} r^2}{c_s^2}
- \frac{12.798946193029\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_2^{c_s} r}{c_s}
- \frac{0.40191656771049\alpha_0^{c_s}\alpha_1^s r^4\log\left(c_s\right)}{c_s^4}
- \frac{0.92232932712305\alpha_0^s\alpha_3^s r^2\log\left(c_s\right)}{c_s^2}
- \frac{0.46845360803068\alpha_0^s\alpha_1^s r^4\log\left(c_s\right)}{c_s^4}
- \frac{0.9424780878218\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2 r^3\log\left(c_s\right)^2}{c_s^3}
- \frac{15.157229920671\alpha_0^{c_s}\alpha_0^s\left(\alpha_1^s\right)^2 r}{c_s}
- \frac{0.26542847151571\alpha_0^s\alpha_1^{c_s} r^4\log\left(c_s\right)}{c_s^4}
- \frac{17.070189849163\alpha_0^s\alpha_1^{c_s}\alpha_1^s r^2}{c_s^2}
- \frac{17.265090188972\alpha_0^{c_s}\alpha_0^s\left(\alpha_1^{c_s}\right)^2 r}{c_s}
$$

$$
- \frac{0.036458333333333 \alpha_0^{c_s} \alpha_3^s r^2 \log\left(c_s\right)^3}{c_s^2}
- \frac{19.139624219519 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_2^s r}{c_s}
- \frac{10.401045538112 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_2^s r}{c_s}
- \frac{0.020833333333333 \alpha_1^{c_s} \alpha_2^s r^2 \log\left(c_s\right)^3}{c_s^2}
- \frac{0.083333333333333 \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r^2 \log\left(c_s\right)^3}{c_s^2}
- \frac{2.8326954049503 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r^3 \log\left(c_s\right)}{c_s^3}
- \frac{1.4228234146217 \alpha_1^s \alpha_2^{c_s} r^2 \log\left(c_s\right)}{c_s^2}
- \frac{2.8877107655697 \left(\alpha_0^{c_s}\right)^2 \alpha_1^s r^3 \log\left(c_s\right)}{c_s^3}
- \frac{0.73958333333333 \left(\alpha_0^s\right)^3 \alpha_1^s r^2 \log\left(c_s\right)^3}{c_s^2}
- \frac{6.1460976449703 \left(\alpha_0^{c_s}\right)^3 \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2}
- \frac{0.19270833333333 \alpha_1^s \alpha_2^s r^2 \log\left(c_s\right)^3}{c_s^2}
- \frac{1.5643173084922 \left(\alpha_0^s\right)^4 \alpha_1^s r \log\left(c_s\right)}{c_s}
- \frac{0.098958333333333 \alpha_0^s \alpha_3^s r^2 \log\left(c_s\right)^3}{c_s^2}
- \frac{1.5884741815718 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 r^3 \log\left(c_s\right)^2}{c_s^3}
- \frac{6.3886016504085 \left(\alpha_0^s\right)^3 \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2}
- \frac{1.6752754126021 \left(\alpha_0^s\right)^2 \alpha_3^s r \log\left(c_s\right)}{c_s}
- \frac{7.0146178742227 \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r^2 \log\left(c_s\right)}{c_s^2}
- \frac{1.7599670334241 \alpha_1^s \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2}
- \frac{0.45560805454892 \alpha_0^s \alpha_3^{c_s} r^2 \log\left(c_s\right)}{c_s^2}
- \frac{14.707067627279 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_2^{c_s} r}{c_s}
- \frac{1.8602805292847 \alpha_1^{c_s} \alpha_2^s r^2 \log\left(c_s\right)}{c_s^2}
- \frac{15.16148696948 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} \alpha_1^s r}{c_s}
- \frac{3.9539880896494 \left(\alpha_0^{c_s}\right)^2 \alpha_3^s r \log\left(c_s\right)}{c_s}
+ \frac{0.125 \alpha_0^s \alpha_1^{c_s} \alpha_1^s r^2 \log\left(c_s\right)^3}{c_s^2}
$$

$$+\frac{0.25\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.375\left(\alpha_0^s\right)^2\alpha_1^{c_s}\alpha_1^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.625\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{0.9375\alpha_0^{c_s}\alpha_0^s\left(\alpha_1^s\right)^2 r\log\left(c_s\right)^3}{c_s}$$
$$+\frac{3.75\alpha_0^s\alpha_1^s\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.56759071136537\alpha_0^{c_s}\alpha_1^s\alpha_2^{c_s} r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{0.85138606704805\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2\alpha_1^{c_s} r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{3.2027721340961\alpha_0^s\alpha_1^{c_s}\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{6.243371063951\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_2^{c_s} r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.6520791005721\alpha_0^s\alpha_1^s\alpha_2^{c_s} r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{1.7483744562548\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^{c_s} r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{1.8395791005721\alpha_0^{c_s}\alpha_1^{c_s}\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{1.8395791005721\alpha_0^{c_s}\left(\alpha_0^s\right)^3\alpha_1^{c_s} r\log\left(c_s\right)^2}{c_s}$$

$$+\frac{2.0270791005721\left(\alpha_0^{c_s}\right)^3\alpha_0^s\alpha_1^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{2.5844883892067\alpha_0^{c_s}\alpha_0^s\alpha_3^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{5.1689767784134\alpha_0^{c_s}\alpha_1^s\alpha_2^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{5.1689767784134\alpha_0^{c_s}\left(\alpha_0^s\right)^3\alpha_1^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{5.9291582011442\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2\alpha_1^s r\log\left(c_s\right)^2}{c_s}$$
$$+\frac{4.8548512346682\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_1^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{10.317828730023\left(\alpha_0^{c_s}\right)^2\alpha_1^{c_s}\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{10.735178932792\alpha_0^s\alpha_1^{c_s}\alpha_1^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{15.682802633666\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_2^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{8.5357704348493\alpha_0^{c_s}\alpha_0^s\alpha_2^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{11.681799284612\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_2^s r\log\left(c_s\right)}{c_s}$$

$$+\frac{3.2060453025497\alpha_0^{c_s}\alpha_0^s\alpha_2^{c_s}r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{14.532737823278\left(\alpha_0^s\right)^2\alpha_1^{c_s}\alpha_1^s r\log\left(c_s\right)}{c_s}$$
$$+\frac{8.1445500735889\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^s r^2\log\left(c_s\right)^2}{c_s^2}$$
$$+\frac{17.422836633219\alpha_0^{c_s}\alpha_0^s\left(\alpha_1^s\right)^2 r\log\left(c_s\right)}{c_s}$$
$$+\frac{8.7476479574518\alpha_0^{c_s}\alpha_1^{c_s}\alpha_1^s r^2\log\left(c_s\right)}{c_s^2}$$
$$+\frac{0.29166666666667\alpha_0^{c_s}\alpha_0^s\alpha_2^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$+\frac{2.4575502603924\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_2^{c_s}r\log\left(c_s\right)}{c_s}$$
$$+\frac{1.5884741815718\alpha_0^{c_s}\alpha_0^s\alpha_1^s r^3\log\left(c_s\right)^2}{c_s^3}$$
$$+\frac{3.4988253905886\alpha_0^{c_s}\alpha_0^s\left(\alpha_1^{c_s}\right)^2 r\log\left(c_s\right)}{c_s}$$
$$-\frac{0.0625\alpha_0^{c_s}\alpha_1^{c_s}\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.0625\alpha_0^{c_s}\left(\alpha_0^s\right)^3\alpha_1^{c_s}r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.0625\alpha_0^s\alpha_1^s\alpha_2^{c_s}r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.0625\left(\alpha_0^{c_s}\right)^3\alpha_0^s\alpha_1^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.25\alpha_0^s\alpha_1^{c_s}\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.375\left(\alpha_0^{c_s}\right)^2\left(\alpha_0^s\right)^2\alpha_1^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.3125\alpha_0^{c_s}\alpha_0^s\alpha_3^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.625\alpha_0^{c_s}\alpha_1^s\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.625\alpha_0^{c_s}\left(\alpha_0^s\right)^3\alpha_1^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{1.25\alpha_0^s\alpha_1^s\alpha_2^s r\log\left(c_s\right)^3}{c_s}$$
$$-\frac{0.21875\left(\alpha_0^{c_s}\right)^2\alpha_0^s\alpha_1^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$-\frac{0.875\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_1^s r^2\log\left(c_s\right)^3}{c_s^2}$$
$$-\frac{0.10546875\alpha_0^{c_s}\alpha_0^s\alpha_1^s r^3\log\left(c_s\right)^3}{c_s^3}$$
$$-\frac{0.56759071136537\alpha_0^{c_s}\left(\alpha_0^s\right)^2\alpha_2^{c_s}r\log\left(c_s\right)^2}{c_s}$$
$$-\frac{11.748374147078\alpha_0^{c_s}\alpha_1^s\alpha_2^s r\log\left(c_s\right)}{c_s}$$

$$- \frac{0.85138606704805 \left(\alpha_0^{c_s}\right)^2 \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{3.9527721340961 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_2^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{6.243371063951 \alpha_0^{c_s} \alpha_1^s \alpha_2^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{0.87418722812738 \alpha_0^{c_s} \alpha_1^{c_s} \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{4.8041582011442 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{5.1689767784134 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_2^s r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{2.3336756173341 \alpha_0^s \alpha_1^{c_s} \alpha_1^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{9.7772583654733 \alpha_0^s \alpha_1^{c_s} \alpha_2^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{10.317828730023 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{10.419919562718 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 \alpha_1^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{2.7829825838101 \alpha_0^{c_s} \alpha_0^s \alpha_2^s r^2 \log\left(c_s\right)^2}{c_s^2}$$
$$- \frac{5.9407619360049 \alpha_0^{c_s} \alpha_0^s \alpha_3^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{49.191233535148 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} \alpha_1^s r}{c_s}$$
$$- \frac{6.5235686838326 \alpha_0^s \alpha_1^s \alpha_2^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{7.7534651676201 \alpha_0^{c_s} \alpha_0^s \left(\alpha_1^s\right)^2 r \log\left(c_s\right)^2}{c_s}$$
$$- \frac{7.8879529823954 \alpha_0^s \alpha_1^s \alpha_2^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{8.876593450377 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^{c_s} r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{10.486494425184 \alpha_0^{c_s} \alpha_1^{c_s} \alpha_2^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{25.076682142992 \left(\alpha_0^{c_s}\right)^2 \alpha_0^s \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{13.746269384684 \left(\alpha_0^{c_s}\right)^3 \alpha_0^s \alpha_1^s r \log\left(c_s\right)}{c_s}$$
$$- \frac{17.495295914904 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^{c_s} r^2 \log\left(c_s\right)}{c_s^2}$$
$$- \frac{2.2075502603924 \left(\alpha_0^{c_s}\right)^3 \alpha_0^s \alpha_1^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{2.5825502603924 \alpha_0^s \alpha_1^{c_s} \alpha_2^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{1.4162751301962 \alpha_0^{c_s} \alpha_0^s \alpha_3^{c_s} r \log\left(c_s\right)}{c_s}$$
$$- \frac{11.54864955968 \alpha_0^{c_s} \left(\alpha_0^s\right)^3 \alpha_1^s r \log\left(c_s\right)}{c_s}$$

$$
\begin{aligned}
&- \frac{23.391054225566 \left(\alpha_0^{c_s}\right)^2 \left(\alpha_0^s\right)^2 \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&- \frac{24.515181114859 \alpha_0^{c_s} \left(\alpha_0^s\right)^2 \alpha_1^s r^2 \log\left(c_s\right)}{c_s^2} \\
&- \frac{3.3676073153919 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} r^3 \log\left(c_s\right)}{c_s^3} \\
&- \frac{6.9540580302476 \alpha_0^{c_s} \alpha_0^s \alpha_1^s r^3 \log\left(c_s\right)}{c_s^3} \\
&+ \frac{0.1875 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^3}{c_s} \\
&+ \frac{31.326333550619 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)}{c_s} \\
&- \frac{5.5187373017162 \alpha_0^{c_s} \alpha_0^s \alpha_1^{c_s} \alpha_1^s r \log\left(c_s\right)^2}{c_s}
\end{aligned}
$$